Sub-Gaussian Concentration

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1 Sub-Gaussian Random Variables

1.1 Moment-Generating Function and Chernoff Bound

In probability theory, the moment-generating function is an alternative characterization of probability distributions. The k-th moment of a distribution can be obtained by evaluating the k-th derivative of its moment-generating function at 0, as is implied by the nomenclature. In contrast to characteristic functions, the moment-generating function of a distribution does not necessarily exist. (For counterexample, consider a standard Cauchy distribution with density $\frac{1}{\pi(1+x^2)}$, $-\infty < x < \infty$.)

Definition 1.1 (Moment-generating function, MGF)**.** Let X be a real-valued random variable such that $\mathbb{E}[e^{tX}]$ exists in some neighborhood of 0, i.e. there exists $\delta > 0$ such that $\mathbb{E}[e^{tX}] < \infty$ for $t \in (-\delta, \delta)$. We define the *moment-generating function (MGF)* of X, denoted by M_X, as

$$M_X(t) := \mathbb{E}[e^{tX}], \quad t \in (-\delta, \delta),$$

and the *centered MGF* as

$$M_X^*(t) := \mathbb{E}[e^{t(X-\mathbb{E}X)}] = e^{-t\mathbb{E}X} M_X(t), \quad t \in (-\delta, \delta).$$

In the above definition the existence of first-moment $\mathbb{E}X$ is ensured by the existence of MGF.

Remark. Note that for all $t \in (0, \delta)$, we have

$$0 \le 1 + \sum_{k=1}^{\infty} \frac{|tX|^k}{k!} = e^{t|X|} \le e^{tX} + e^{-tX}.$$

Therefore $\mathbb{E}|X|^k < \infty$ for all $k \in \mathbb{N}$. Furthermore, the series $\sum_{k=0}^{\infty} \mathbb{E}[X^k]t^k/k!$ is absolutely convergent, and

$$\left| M_X(t) - \sum_{k=0}^{N} \frac{\mathbb{E}[X^k]}{k!} t^k \right| \le \sum_{k=N+1}^{\infty} \frac{\mathbb{E}|X|^k}{k!} t^k \to 0 \quad \text{as } N \to \infty.$$

Therefore, $M_X(t)$ has the following power series representation:

$$M_X(t) = \sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!} t^k, \quad t \in (-\delta, \delta).$$

Indeed, the function $\mathbb{C} \to \mathbb{C} : z \mapsto \mathbb{E}[e^{zX}]$ is holomorphic in the strip $\{z \in \mathbb{C} : -\delta < \Re(z) < \delta\}$ and admits the above representation. Also, we pick $\epsilon > 0$ such that $(t-\epsilon, t+\epsilon) \subset (0, \delta)$. Then

$$|X|^k e^{tX} \le \frac{k!}{\epsilon^k} e^{\epsilon|X|} e^{tX} \le \frac{k!}{\epsilon^k} \left(e^{(t-\epsilon)X} + e^{(t+\epsilon)X} \right),$$

and $X^k e^{tX}$ is integrable. Hence by Fubini's theorem,

$$\frac{\mathrm{d}^k}{\mathrm{d}t^k} M_X(t) = \sum_{n=k}^{\infty} \frac{\mathbb{E}[X^n]}{(n-k)!} t^{n-k} = \mathbb{E}\left[\sum_{n=1}^{\infty} \frac{X^{n+k}}{n!} t^n \right] = \mathbb{E}[X^k e^{tX}], \quad k = 1, 2, \cdots.$$

Hence we conclude that M_X is infinitely differentiable in $(-\delta, \delta)$. In particular, the moments of X can be recovered from its MGF by $\mathbb{E}[X^k] = M_X^{(k)}(0)$.

In practical situations, we may wonder if our sample is concentrated near the expectation of the population. Applying Markov's inequality to the integrand in MGF, we can attain the Chernoff bound.

Lemma 1.2 (Chernoff bound)**.** *Suppose that $M_X(t) < \infty$ for all $t \in \mathbb{R}$. Then for all $\epsilon, t \geq 0$, we have*

$$\mathbb{P}(X - \mathbb{E}X \geq \epsilon) \leq M_X^*(t)e^{-t\epsilon}.$$

To obtain a tighter bound, take the infimum of RHS:

$$\mathbb{P}(X - \mathbb{E}X \geq \epsilon) \leq \inf_{t \geq 0} M_X^*(t)e^{-t\epsilon}.$$

Following is an example of the Chernoff bound.

Example 1.3 (Gaussian MGF)**.** We look at a simple example focusing on the Chernoff bound of a Gaussian variable $Z \sim N(0, \sigma^2)$. The MGF of variable Z is

$$M_Z^*(t) = \frac{1}{\sqrt{2\pi}\sigma} \int e^{tz - \frac{z^2}{2\sigma^2}} \mathrm{d}z = \exp\left(\frac{t^2\sigma^2}{2}\right).$$

And we get the Gaussian tail bound

$$\mathbb{P}(Z \geq \epsilon) \leq \inf_{t \geq 0} \exp\left(\frac{t^2\sigma^2}{2} - t\epsilon\right) = \exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \tag{1.1}$$

We can extend this tail bound to a more general class of random variables.

1.2 Sub-Gaussian Random Variables

Following Example 1.3, we can further conclude that if a random variable X satisfies $M_X^*(t) \leq \exp\left(t^2\sigma^2/2\right)$ for all $t \in \mathbb{R}$, then a decay rate in form of (1.1) can be obtained. This motivates the definition of sub-Gaussian random variables.

Definition 1.4 (Sub-Gaussian random variable)**.** Let $\sigma > 0$. A random variable X with mean $\mu = \mathbb{E}X$ is said to be *sub-Gaussian with variance proxy* σ^2 (or σ^2*-sub-Gaussian*), written $X \in \mathrm{SG}(\sigma^2)$, if

$$M_X^*(t) = \mathbb{E}\left[e^{t(X-\mu)}\right] \leq \exp\left(\frac{t^2\sigma^2}{2}\right), \quad \text{for all } t \in \mathbb{R}.$$

By definition, any σ^2-sub-Gaussian random variable is also ρ^2-sub-Gaussian for each $\rho > \sigma$. This definition generalizes Gaussian tail bounds (1.4) to non-Gaussian variables satisfying the MGF condition. There are several equivalent definitions of sub-Gaussianity. Following is an exercise in Handel's book [2], Chapter 3.

Theorem 1.5 (Characterizations of sub-Gaussianity)**.** *Let X be a centered random variable, i.e., $\mathbb{E}X = 0$. Then the following statements are equivalent:*

(i) (MGF condition). X is sub-Gaussian.

(ii) (Tail bound condition). There is a constant $\tau > 0$ such that

$$\mathbb{P}(|X| \geq \epsilon) \leq 2\exp\left(-\frac{\epsilon^2}{2\tau^2}\right), \quad \textit{for every } \epsilon > 0. \tag{1.2}$$

(iii) There is a constant $\nu > 0$ such that

$$\mathbb{E}\left[\exp\left(\frac{X^2}{2\nu^2}\right)\right] \leq 2. \tag{1.3}$$

(iv) (Moments of even order) There is a constant $\theta > 0$ such that

$$\mathbb{E}[X^{2k}] \leq \frac{(2k)!}{2^k k!}\theta^{2k}, \quad \textit{for every } k \in \mathbb{N}. \tag{1.4}$$

Proof. (i) $\Rightarrow$ (ii): Let X be σ^2-sub-Gaussian. Fix $\epsilon > 0$. For every $t > 0$, we have

$$\mathbb{P}(X \geq \epsilon) = \mathbb{P}\left(e^{tX} \geq e^{-t\epsilon}\right) \leq e^{t\epsilon}\mathbb{E}[e^{tX}] \leq \exp\left(\frac{1}{2}t^2\sigma^2 - t\epsilon\right).$$

Setting $t = \epsilon/\sigma^2$ implies $\mathbb{P}(X \geq \epsilon) \leq \exp\left(-\epsilon^2/2\sigma^2\right)$. By applying similar calculation to $-X$, we can obtain $\mathbb{P}(X \leq -\epsilon) \leq \exp\left(-\epsilon^2/2\sigma^2\right)$, and the result (1.2) immediately follows for $\tau = \sigma$.

(ii) $\Rightarrow$ (iii): Suppose (1.2) holds for $\tau > 0$. We will use the following fact, if Y is an random variable that is almost surely non-negative and has distribution F, and ϕ is a differentiable increasing function, then

$$\begin{aligned}\mathbb{E}[\phi(Y)] &= \int_0^\infty \phi(y)\mathrm{d}F(y) = \int_0^\infty \left(\phi(0) + \int_0^y \phi'(\epsilon)\mathrm{d}\epsilon\right)\mathrm{d}F(y) \\ &= \phi(0) + \int_0^\infty \int_t^\infty \phi'(\epsilon)\mathrm{d}F(y)\mathrm{d}\epsilon = \phi(0) + \int_0^\infty \mathbb{P}(Y \geq \epsilon)\phi'(\epsilon)\,\mathrm{d}\epsilon.\end{aligned} \tag{1.5}$$

Set $Y = X^2$. For any $\nu > \tau$, we have

$$\begin{aligned}\mathbb{E}\left[\exp\left(\frac{X^2}{2\nu^2}\right)\right] &= 1 + \int_0^\infty \frac{1}{2\nu^2}\exp\left(\frac{\epsilon}{2\nu^2}\right)\mathbb{P}(X^2 \geq \epsilon)\,\mathrm{d}\epsilon \\ &\leq 1 + \frac{1}{\nu^2}\int_0^\infty \exp\left[\epsilon\left(\frac{1}{2\nu^2} - \frac{1}{2\tau^2}\right)\right]\mathrm{d}\epsilon = 1 + \frac{2\tau^2}{\nu^2 - \tau^2},\end{aligned} \tag{1.6}$$

where the inequality follows from (1.2). Then we can attain (1.3) by setting $\nu = \sqrt{3}\tau$ in (1.6).

(iii) $\Rightarrow$ (iv): Note that $e^x \geq 1 + x^k/k!$ for $x \geq 0$ and $k \in \mathbb{N}$, we have

$$2 \geq \mathbb{E}\left[\exp\left(\frac{X^2}{2\nu^2}\right)\right] \geq 1 + \frac{\mathbb{E}[X^{2k}]}{k!(2\nu)^k}.$$

Then

$$\mathbb{E}[X^{2k}] \leq k!(2\nu^2)^k \leq \frac{(2k)!}{2^k k!}(2k+1)\nu^{2k},$$

and (1.4) follows for $\theta^2 = 3\nu^2$.

(iv) $\Rightarrow$ (i): Let X' be an independent copy of X and create $Y := X - X'$. Then Y is symmetric, and the odd moments vanish:

$$\mathbb{E}[e^{tY}] = \sum_{k=0}^\infty \frac{t^k\mathbb{E}[Y^k]}{k!} = \sum_{k=0}^\infty \frac{t^{2k}\mathbb{E}[Y^{2k}]}{(2k)!}. \tag{1.7}$$

For any $k \in \mathbb{N}$, we have the c_r-inequality:

$$\mathbb{E}[Y^{2k}] = \mathbb{E}[(X - X')^{2k}] = 2^{2k}\mathbb{E}\left[\left(\frac{X}{2} + \frac{-X'}{2}\right)^{2k}\right] \leq 2^{2k}\mathbb{E}\left[\frac{1}{2}X^{2k} + \frac{1}{2}(-X')^{2k}\right] = 2^{2k}\mathbb{E}[X^{2k}].$$

Plug in to (1.7), we have

$$\mathbb{E}[e^{tY}] \leq \sum_{k=0}^\infty \frac{(2t)^{2k}\mathbb{E}[X^{2k}]}{(2k)!} \leq \sum_{k=0}^\infty \frac{2^k(t\theta)^{2k}}{k!} \leq \exp\left(2t^2\theta^2\right).$$

Since $\mathbb{E}X = 0$, and by Jensen's inequality,

$$\mathbb{E}[e^{tX}] = \mathbb{E}[e^{tX - t\mathbb{E}X'}] \leq \mathbb{E}[e^{t(X-X')}] = \mathbb{E}[e^{tY}] \leq \exp\left(2t^2\theta^2\right).$$

Then X is sub-Gaussian with proxy $\sigma^2 = 4\theta^2$. $\square$

We further introduce the concept of sub-Gaussian vectors.

Definition 1.6 (Sub-Gaussian vector)**.** Let $\sigma > 0$. A random vector $X = (X_1, \cdots, X_n)^\top$ is said to be a *σ^2-sub-Gaussian vector* , if for all $u \in \mathbb{R}^n$ with $\|u\|_2 = 1$, we have $u^\top X \in \mathrm{SG}(\sigma^2)$.

Proposition 1.7. *Suppose $X_1, \cdots, X_n$ are independent sub-Gaussian variables with variance proxy σ^2. Then $X = (X_1, \cdots, X_n)^\top$ is a σ^2-sub-Gaussian vector.*

Proof. By direct calculation

$$\mathbb{E}\left[e^{tX^\top u}\right] = \prod_{i=1}^n \mathbb{E}\left[e^{tu_i X_i}\right] \le \prod_{i=1}^n \exp\left(\frac{t^2 u_i^2 \sigma^2}{2}\right) = \exp\left(\frac{t^2\sigma^2\|u\|^2}{2}\right) = \exp\left(\frac{t^2\sigma^2}{2}\right). \qquad \square$$

The sub-Gaussian family contains a very wide range of random variables, such as Gaussian variables, Rademacher variables and bounded variables. We will see some illustrative examples below.

Proposition 1.8 (Rademacher variables are sub-Gaussian)**.** *Let X be a Rademacher random variable, i.e. $\mathbb{P}(X = 1) = \mathbb{P}(X = -1) = 1/2$. Then $X \in \mathrm{SG}(1)$.*

Proof. For all $t \in \mathbb{R}$,

$$\mathbb{E}[e^{tX}] = \frac{e^t + e^{-t}}{2} = \sum_{k=0}^\infty \frac{t^{2k}}{(2k)!} \le \sum_{k=0}^\infty \frac{t^{2k}}{2^k k!} = e^{t^2/2}. \qquad \square$$

Lemma 1.9 (Hoeffding's lemma)**.** *Fix $a < b$. Suppose X is a random variable such that $\mathbb{P}(X \in [a, b]) = 1$. Then $X \in \mathrm{SG}((b-a)^2/4)$.*

Proof. This proof is adapted from Handel's notes. Without loss of generality, let $\mathbb{E}X = 0$. Use exponential tilting. Fix $t \in \mathbb{R}$. For any Borel set $B \in \mathcal{B}(\mathbb{R})$, define $\mathbb{P}_t : \mathcal{B}(\mathbb{R}) \to \mathbb{R}$ as

$$\mathbb{P}_t(B) := \frac{\mathbb{E}\left[e^{tX}\mathbb{1}_{\{X\in B\}}\right]}{\mathbb{E}[e^{tX}]}.$$

It can be verified that $\mathbb{P}_t$ is a valid probability measure on $\mathbb{R}$. Let random variable $U_t \sim \mathbb{P}_t$. Using simple approximation theorem, we have for any measurable function f that

$$\mathbb{E}[f(U_t)] = \frac{\mathbb{E}[e^{tX} f(X)]}{\mathbb{E}[e^{tX}]}.$$

Now we investigate the logarithmic MGF $\psi_X(t) = \log \mathbb{E}[e^{tX}]$. Using the interchangeability of derivative and integral (implied by the dominated convergence theorem), we have

$$\psi_X'(t) = \frac{\mathbb{E}[Xe^{tX}]}{\mathbb{E}[e^{tX}]} = \mathbb{E}[U_t], \qquad \psi_X''(t) = \frac{\mathbb{E}[X^2 e^{tX}]}{\mathbb{E}[e^{tX}]} - \left(\frac{\mathbb{E}[Xe^{tX}]}{\mathbb{E}[e^{tX}]}\right)^2 = \mathrm{Var}(U_t). \tag{1.8}$$

By definition, $\mathbb{P}(U_t \in [a, b]) = \mathbb{P}_t([a, b]) = 1$, hence

$$\mathrm{Var}(U_t) = \mathbb{E}[(U_t - \mathbb{E}U_t)^2] = \inf_{c\in\mathbb{R}} \mathbb{E}[(U_t - c)^2] \le \mathbb{E}\left[\left(U_t - \frac{a+b}{2}\right)^2\right] \le \left(\frac{b-a}{2}\right)^2. \tag{1.9}$$

Using (1.8) and (1.9), we can bound ψ_X as follows:

$$\psi_X(t) = \psi_X(0) + \int_0^t \left(\psi_X'(0) + \int_0^s \psi_X''(u)\mathrm{d}u\right)\mathrm{d}s \le \int_0^t \int_0^s \left(\frac{b-a}{2}\right)^2 \mathrm{d}u\,\mathrm{d}s = \frac{t^2(b-a)^2}{8}.$$

Thus we complete the proof. $\square$

1.3 Sub-Exponential Variables

The class of sub-exponential random variables can be viewed as a natural extension of the class of sub-Gaussian random variables.

Definition 1.10. Let $\nu, \alpha > 0$. A random variable X with mean $\mu = \mathbb{E}X$ is said to be *sub-exponential with parameters* $\nu, \alpha > 0$, written $X \sim \mathrm{SE}(\nu^2, \alpha)$, if

$$M_X^*(t) = \mathbb{E}\left[e^{t(X-\mu)}\right] \le \exp\left(\frac{t^2\nu^2}{2}\right), \quad \text{for all } t \in \left(-\frac{1}{\alpha}, \frac{1}{\alpha}\right).$$

By definition, any $X \in \mathrm{SG}(\sigma^2)$ is also a sub-exponential random variable $X \in \mathrm{SE}(\sigma^2, \alpha)$ for every $\alpha > 0$. Following are some equivalent characterizations of sub-exponential variables.

Theorem 1.11 (Characterization of sub-exponentiality)**.** *Let X be a centered random variable, i.e. $\mathbb{E}X = 0$. Then the following statements are equivalent:*

(i) (MGF condition). X is sub-exponential;

(ii) (Tail bound condition). There exists constants $\nu, \alpha > 0$ such that

$$\mathbb{P}(|X| \ge \epsilon) \le 2\exp\left(-\frac{1}{2}\min\left\{\frac{\epsilon^2}{\nu^2}, \frac{\epsilon}{\alpha}\right\}\right), \quad \textit{for every } \epsilon > 0.$$

(iii) There exists a constant $\gamma > 0$ such that

$$\mathbb{E}[e^{|X|/\gamma}] \le 2.$$

(iv) (Bernstein's condition) The second moment $\sigma^2 = \mathbb{E}|X|^2 < \infty$, and there exists a constant $b > 0$ such that

$$\mathbb{E}|X|^k \le \frac{k!}{2}\sigma^2 b^{k-2} \quad \textit{for } k = 3, 4, \cdots. \tag{1.10}$$

Proof. (i) $\Rightarrow$ (ii). Let $X \in \mathrm{SE}(\nu^2, \alpha)$ with $\mathbb{E}X = 0$. Using the Chernoff bound, we have

$$\mathbb{P}(X \ge \epsilon) \le \mathbb{P}(e^{tX} \ge e^{t\epsilon}) \le \exp\left(-t\epsilon + \frac{t^2\nu^2}{2}\right) \quad \text{for } |t| < \frac{1}{\alpha}.$$

By direct calculation,

$$\inf_{t\in[0,1/\alpha)}\left(-t\epsilon + \frac{t^2\nu^2}{2}\right) = \begin{cases} -\frac{\epsilon^2}{2\nu^2}, & \epsilon \le \frac{\nu^2}{\alpha}, \\ -\frac{\epsilon}{\alpha} + \frac{\nu^2}{2\alpha^2}, & \epsilon > \frac{\nu^2}{\alpha}. \end{cases}$$

Also we have $-\epsilon/\alpha + \nu^2/2\alpha^2 \le -\epsilon/2\alpha$ when $\epsilon > \nu^2/\alpha$. Then we have (ii):

$$\mathbb{P}(|X| \ge \epsilon) \le \begin{cases} 2\exp\left(-\frac{\epsilon^2}{2\nu^2}\right), & 0 < \epsilon \le \frac{\nu^2}{\alpha}, \\ 2\exp\left(-\frac{\epsilon}{2\alpha}\right), & \epsilon > \frac{\nu^2}{\alpha}. \end{cases}$$

(ii) $\Rightarrow$ (iii). For every $t < 1/\alpha$,

$$\begin{aligned}\mathbb{E}\left[e^{t|X|}\right] &= 1 + \int_0^\infty te^{tx}\mathbb{P}(|X| \ge x)\,\mathrm{d}x = 1 + 2t\left(\int_0^{\nu^2/\alpha} e^{tx - \frac{x^2}{2\nu^2}}\,dx + \int_{\nu^2/\alpha}^\infty e^{(t-1/\alpha)x}\,\mathrm{d}x\right) \\ &\le 1 + 2t\left(\int_0^{\nu^2/\alpha} e^{tx}\,dx + \int_0^\infty e^{(t-1/\alpha)x}\,\mathrm{d}x\right) = 1 + 2(e^{t\nu^2/\alpha} - 1) + \frac{2\alpha t}{1-\alpha t}.\end{aligned}$$

If $t = \min\{\alpha\nu^{-2}\log(5/4), 1/(5\alpha)\}$, we have $\mathbb{E}[e^{t|X|}] \le 2$. It suffices to choose $\gamma = 1/t$ in (iii).

(iii) $\Rightarrow$ (iv). Note that $e^x \geq 1 + x^k/k!$ for $x \geq 0$ and $k \in \mathbb{N}$. Then we have

$$\sigma^2 = \mathbb{E}|X|^2 \leq 2\gamma^2 \left(\mathbb{E}\big[e^{|X|/\gamma}\big] - 1\right) \leq 2\gamma^2,$$

and

$$\mathbb{E}|X|^k \leq k!\gamma^k \left(\mathbb{E}\big[e^{|X|/\gamma}\big] - 1\right) \leq k!\gamma^k \leq \frac{k!}{2}\sigma^2\gamma^{k-2}\left(\frac{2\gamma^2}{\sigma^2}\right) \leq \frac{k!}{2}\sigma^2\gamma^{k-2}\left(\frac{2\gamma^2}{\sigma^2}\right)^{k-2}.$$

Then (iv) holds for $b = 2\gamma^3/\sigma^2$.

(iv) $\Rightarrow$ (i). For $t \in (-1/b, 1/b)$, by Tonelli's theorem,

$$\begin{aligned}\mathbb{E}\big[e^{|tX|}\big] &= 1 + \frac{|t|^2\sigma^2}{2} + \sum_{k=3}^{\infty}\frac{|t|^k\mathbb{E}|X|^k}{k!} \\ &\leq 1 + \frac{\sigma^2}{2}\left(|t|^2 + \sum_{k=3}^{\infty}|t|^k b^{k-2}\right) = 1 + \frac{|t|^2\sigma^2}{2(1-b|t|)} \leq \exp\left(\frac{t^2\sigma^2}{2(1-b|t|)}\right). \end{aligned} \tag{1.11}$$

Taking $t \in (-1/2b, 1/2b)$, the bound becomes

$$\mathbb{E}\big[e^{|tX|}\big] \leq \exp\left(\frac{t^2\sigma^2}{2(1-b|t|)}\right) \leq e^{t^2\sigma^2}.$$

Note that $\mathbb{E}[e^{tX}] \leq \mathbb{E}[e^{|tX|}]$. Therefore $X \in \mathrm{SE}(2\sigma^2, 2b)$, and we complete the proof. □

Corollary 1.12 (Bernstein's inequality)**.** *Let X be a random variable with mean μ and variance σ^2. If X satisfies the Bernstein's condition* (1.10)*, then*

$$\mathbb{P}(|X-\mu| \geq \epsilon) \leq 2\exp\left(-\frac{\epsilon^2}{2(\sigma^2 + b\epsilon)}\right).$$

Proof. We may assume $\mu = 0$. Let $t \in (0, 1/b)$. Following (1.11), we have

$$\mathbb{E}\big[e^{tX}\big] \leq \exp\left(\frac{t^2\sigma^2}{2(1-bt)}\right), \tag{1.12}$$

and use the Chernoff bound to obtain

$$\mathbb{P}(X \geq \epsilon) \leq e^{-t\epsilon}\mathbb{E}\left[e^{tX}\right] \leq \exp\left(-t\epsilon + \frac{t^2\sigma^2}{2(1-bt)}\right).$$

Setting $t = \epsilon/(\sigma^2 + b\epsilon)$ gives

$$\mathbb{P}(X \geq \epsilon) \leq \exp\left(-\frac{\epsilon^2}{2(\sigma^2 + b\epsilon)}\right).$$

A similar results also holds for sub-exponential variable $-X$. Then (1.12) holds by taking a union bound. □

For sub-exponential variables, we also have a composition property similar to Proposition 1.7.

Proposition 1.13. *Let $X_i \in \mathrm{SE}(\nu_i^2, \alpha_i)$, $i = 1, \cdots, n$ be independent sub-exponential variables. Then the sum $X_1 + \cdots + X_n$ is sub-exponential with parameters $\nu^2 = \nu_1^2 + \cdots + \nu_n^2$ and $\alpha = \max_{1\leq i\leq n}\alpha$.*

Proof. Fix $|t| < \min_{1\leq i\leq n} 1/\alpha_i$. Then

$$\mathbb{E}\left[e^{t(X_1+\cdots+X_n)}\right] = \prod_{i=1}^{n}\mathbb{E}\left[e^{tX_i}\right] \leq \prod_{i=1}^{n}\exp\left(\frac{t^2\nu_i^2}{2}\right) = \exp\left(\frac{t^2}{2}(\nu_1^2 + \cdots + \nu_n^2)\right).$$

Thenwe finish the proof. □

Example 1.14 (Bounded random variables)**.** Let X be a bounded variable such that $|X - \mathbb{E}X| \le b$ a.s.. Then X satisfies the Bernstein condition

$$\mathbb{E}|X|^k \le \frac{k!}{2}\mathbb{E}|X|^2 \left(\frac{b}{3}\right)^{k-2} \quad \text{for } k = 3, 4, \cdots.$$

Then $X \in \mathrm{SE}(2\mathbb{E}|X|^2, 2b/3)$. Furthermore, we have the Bernstein's inequality

$$\mathbb{P}\left(|X - \mu| \ge \epsilon\right) \le 2\exp\left(-\frac{\epsilon^2}{2(\mathbb{E}|X|^2 + b\epsilon/3)}\right). \tag{1.13}$$

Example 1.15 (Squared sub-Gaussian variables)**.** Let X be a sub-Gaussian random variable with variance proxy $\sigma^2 > 0$. Then $Z = X^2 - \mathbb{E}[X^2]$ is sub-exponential. To see this, note that for $k \in \mathbb{N}$,

$$\mathbb{E}[X^{2k}] = \int_0^\infty kt^{k-1}\mathbb{P}(X^2 \ge t)\,\mathrm{d}t \le 2\int_0^\infty kt^{k-1}\exp\left(-\frac{t}{2\sigma^2}\right)\mathrm{d}t = 2k(2\sigma^2)^k\Gamma(k) = 2^{k+1}k!\sigma^{2k}.$$

Since $|X^2 - \mathbb{E}[X^2]|^k \le \max\{X^{2k}, (\mathbb{E}[X^2])^k\} \le X^{2k} + (\mathbb{E}[X^2])^k$, and by the above estimate, we have

$$\begin{aligned}
\mathbb{E}\left[e^{tZ}\right] &= 1 + \sum_{k=2}^\infty \frac{t^k\mathbb{E}\left[(X^2 - \mathbb{E}[X^2])^k\right]}{k!} \le 1 + \sum_{k=2}^\infty \frac{t^k\mathbb{E}\left[\max\{X^{2k}, (\mathbb{E}[X^2])^k\}\right]}{k!} \\
&\le 1 + \sum_{k=2}^\infty \frac{2t^k\mathbb{E}\left[X^{2k}\right]}{k!} \le 1 + \sum_{k=2}^\infty 2^{k+2}t^k\sigma^{2k} \\
&= 1 + \frac{16t^2\sigma^4}{1 - 2t\sigma^2} \le 1 + 32t^2\sigma^4, \quad \text{for } |t| \le \frac{1}{4\sigma^2}.
\end{aligned}$$

Hence $Z \sim \mathrm{SE}(64\sigma^4, 4\sigma^2)$. Furthermore, we have the tail bound

$$\mathbb{P}\left(|X| \ge \epsilon\right) \le 2\exp\left(-\frac{1}{2}\min\left\{\frac{\epsilon^2}{64\sigma^4}, \frac{\epsilon}{4\sigma^2}\right\}\right), \quad \epsilon > 0.$$

Following are some non-sub-Gaussian examples.

Example 1.16 (Gamma distribution family)**.** Let X have the Gamma distribution with shape $\alpha > 0$ and scale $\theta > 0$, i.e.

$$X \sim \frac{x^{\alpha-1}}{\theta^\alpha\Gamma(\alpha)}e^{-x/\theta}\mathbb{1}_{\{x\ge 0\}}\,\mathrm{d}x.$$

Then X is sub-exponential, and $X \in \mathrm{SE}(2\alpha\theta^2, 2\theta)$. In particular,

(i) if $X \sim \mathrm{Exponential}(\lambda)$ has the exponential distribution with rate $\lambda > 0$, then $X \in \mathrm{SE}(2/\lambda^2, 2/\lambda)$;

(ii) if $X \sim \chi^2(n)$ has the chi-square distribution with degree of freedom $n \ge 1$, then $X \in \mathrm{SE}(4n, 4)$.

Proof. By direct calculation we have $\mathbb{E}[X] = \alpha\theta$. For $|t| < 1/\theta$,

$$\mathbb{E}[e^{tX}] = \int_0^\infty \frac{x^{\alpha-1}}{\theta^\alpha\Gamma(\alpha)}e^{-x/\theta}e^{tx}\,\mathrm{d}x = \left(\frac{1}{1 - \theta t}\right)^\alpha.$$

Note that $e^{-x} \le (1 - x)\,e^{x^2}$ for $x \in [-1/2, 1/2]$. Then for $|t| \le \theta^{-1}/2$,

$$M_X^*(t) = \mathbb{E}[e^{t(X-\alpha\theta)}] = \left(\frac{e^{-\theta t}}{1 - \theta t}\right)^\alpha \le e^{\alpha\theta^2t^2}.$$

Hence $Y \in \mathrm{SE}(2\alpha\theta^2, 2\theta)$. For the two particular cases, note that $\mathrm{Exponential}(\lambda)$ is the Gamma distribution with shape 1 and scale $1/\lambda$, and $\chi^2(n)$ is the Gamma distribution with shape $n/2$ and scale 2. $\square$

Example 1.17 (Poisson distribution)**.** Let $X \sim \text{Poisson}(\lambda)$, where $\lambda > 0$. For every $t \in \mathbb{R}$,

$$\mathbb{E}[e^{tX}] = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!} e^{tk} = e^{\lambda(e^t - 1)}$$

Note that $e^t \leq 1 + t + \frac{t^2}{2} e^{|t|}$. Then

$$\mathbb{E}\left[e^{t(X-\lambda)}\right] = \exp\left[\lambda(e^t - t - 1)\right] \leq \exp\left(\lambda e^{|t|} \frac{t^2}{2}\right).$$

Hence for any $\alpha > 0$, we have $X \in \text{SE}(\lambda e^{1/\alpha}, \alpha)$.

Example 1.18 (Geometric distribution)**.** Let $X \sim \text{Geo}(\theta)$, where $0 < \theta < 1$. For every $t \leq -\log(1-\theta)$,

$$\mathbb{E}[e^{tX}] = \sum_{k=1}^{\infty} (1-\theta)^{k-1} \theta e^{tk} = \frac{\theta e^t}{1 - (1-\theta)e^t} = \frac{1}{1 - \frac{e^t - 1}{\theta e^t}}.$$

Note that $1 + x \leq e^x \leq 1 + x + \frac{x^2}{2} e^{|x|}$. Then for $|t| \leq \theta/2$, we have

$$\mathbb{E}[e^{t(X-1/\theta)}] = \frac{e^{-t/\theta}}{1 - (1 - e^{-t})/\theta} \leq \frac{1 - \frac{t}{\theta} + \frac{t^2}{2\theta^2} e^{|t|/\theta}}{1 - \frac{t}{\theta}} = 1 + \frac{t^2}{2\theta^2} \frac{e^{|t|/\theta}}{1 - \frac{t}{\theta}} \leq 1 + \frac{t^2 \sqrt{e}}{\theta^2} \leq e^{\sqrt{e} t^2/\theta^2}.$$

Hence $X \in \text{SE}(2\sqrt{e}/\theta^2, 2/\theta)$.

1.4 Orlicz Norms

In this subsection we introduce a framework unifying sub-Gaussian and sub-exponential variables. The following definition is motivated by Theorem 1.5 (iii) and Theorem 1.11 (iii).

Theorem 1.19 (Orlicz Norms)**.** *Let $\psi : [0, \infty) \to [0, \infty)$ be a strictly increasing and convex function with $\psi(0) = 0$. The **Orlicz norm** of a random variable X is defined as*

$$\|X\|_\psi = \inf\left\{K > 0 : \mathbb{E}\psi\left(\frac{|X|}{K}\right) \leq 1\right\}.$$

Then $\|\cdot\|_\psi$ is a norm on the vector space of random variables X with $\|X\|_\psi < \infty$. Furthremore, if X satisfies $0 < \|X\|_\psi < \infty$, we have $\mathbb{E}\psi(|X|/\|X\|_\psi) \leq 1$.

Proof. We first verify that $\|\cdot\|_\psi$ is a norm.

(i) It is clear that $\|0\|_\psi = 0$. Conversely, if $\|X\|_\psi = 0$, then $\mathbb{E}\psi(|X|/K) \leq 1$ for all $K > 0$. By Jensen's inequality, $\mathbb{E}\psi(|X|/K) \geq \psi(\mathbb{E}|X|/K)$. We claim that $\psi(t) \uparrow \infty$ as $t \uparrow \infty$. For every $t > 1$, we have

$$0 < \psi(1) \leq \frac{t-1}{t}\psi(0) + \frac{1}{t}\psi(t) = \frac{\psi(t)}{t}.$$

Hence $\psi(t) \geq t\psi(1)$, which goes to ∞ as $t \uparrow \infty$. Next, we take $\alpha > 0$ with $\psi(\alpha) > 1$. If $\mathbb{E}|X| > 0$, the value $K = \mathbb{E}|X|/\alpha$ satisfies $\mathbb{E}\psi(|X|/K) \geq \psi(\mathbb{E}|X|/K) = \psi(\alpha) > 1$, a contradiction. Therefore $\mathbb{E}|X| = 0$, and $X = 0$.

(ii) It follows easily from definition that $\|\alpha X\|_\psi = |\alpha| \, \|X\|_\psi$ for $\alpha \in \mathbb{R}$.

(iii) We take random variables X and Y with finite Orlicz norms. For any $\epsilon > 0$,

$$\begin{aligned}
\mathbb{E}\psi\left(\frac{|X+Y|}{\|X\|_\psi + \|Y\|_\psi + 2\epsilon}\right) &\le \mathbb{E}\psi\left(\frac{|X|+|Y|}{\|X\|_\psi + \|Y\|_\psi + 2\epsilon}\right) \\
&= \mathbb{E}\psi\left(\frac{\|X\|_\psi + \epsilon}{\|X\|_\psi + \|Y\|_\psi + 2\epsilon} \cdot \frac{|X|}{\|X\|_\psi + \epsilon} + \frac{\|Y\|_\psi + \epsilon}{\|X\|_\psi + \|Y\|_\psi + 2\epsilon} \cdot \frac{|Y|}{\|Y\|_\psi + \epsilon}\right) \\
&\le \frac{\|X\|_\psi + \epsilon}{\|X\|_\psi + \|Y\|_\psi + 2\epsilon}\mathbb{E}\psi\left(\frac{|X|}{\|X\|_\psi + \epsilon}\right) + \frac{\|Y\|_\psi + \epsilon}{\|X\|_\psi + \|Y\|_\psi + 2\epsilon}\mathbb{E}\psi\left(\frac{|Y|}{\|Y\|_\psi + \epsilon}\right) \\
&\le \frac{\|X\|_\psi + \epsilon}{\|X\|_\psi + \|Y\|_\psi + 2\epsilon} + \frac{\|Y\|_\psi + \epsilon}{\|X\|_\psi + \|Y\|_\psi + 2\epsilon} = 1.
\end{aligned}$$

Hence $\|X\|_\psi + \|Y\|_\psi + 2\epsilon \ge \|X+Y\|_\psi$. Since $\epsilon > 0$ us arbitrary, we have $\|X\|_\psi + \|Y\|_\psi \ge \|X+Y\|_\psi$.

Now we assume $0 < \|X\|_\psi < \infty$. Then by the monotone convergence theorem,

$$\mathbb{E}\psi\left(\frac{|X|}{\|X\|_\psi}\right) = \lim_{\epsilon \downarrow 0} \mathbb{E}\psi\left(\frac{|X|}{\|X\|_\psi + \epsilon}\right) \le 1.$$

Thus we finish the proof. □

Remark. The Orlicz norms cover a wide range of commonly used norms.

(i) If $\psi(x) = |x|^p$, where $1 \le p < \infty$, the Orlicz norm $\|\cdot\|_\psi$ is the L^p norm.

(ii) for every $\alpha \ge 1$, we let $\psi_\alpha(t) = \exp(t^\alpha) - 1$. Then

$$\|X\|_{\psi_\alpha} = \inf\left\{K > 0 : \mathbb{E}\left[\exp\left(\frac{|X|^\alpha}{K^\alpha}\right)\right] \le 2\right\}.$$

- When $\alpha = 1$, the norm $\|\cdot\|_{\psi_1}$ is the sub-exponential norm.
- When $\alpha = 2$, the norm $\|\cdot\|_{\psi_2}$ is the sub-Gaussian norm.

We then propose a generalized version of Theorems 1.5 and 1.11.

Proposition 1.20. *Let X be a real-valued random variable, and $\alpha \ge 1$. The following are equivalent:*

(i) (Finite $\|\cdot\|_{\psi_\alpha}$ norm). There exists $K > 0$ such that

$$\mathbb{E}\left[\exp\left(\frac{|X|^\alpha}{K^\alpha}\right)\right] \le 2;$$

(ii) (Tail probability). There exists a constant $c > 0$ such that for all $x > 0$,

$$\mathbb{P}(|X| \ge x) \le 2e^{-cx^\alpha};$$

(iii) (Moment growth rate) There exists a constant $\omega > 0$ such that for all $p \ge 1$,

$$(\mathbb{E}|X|^p)^{1/p} \le \omega p^{1/\alpha}.$$

Proof. (i) ⇒ (ii). By the Chernoff bound,

$$\mathbb{P}(|X| \ge x) = \mathbb{P}\left(e^{|X|^\alpha/K^\alpha} \ge e^{x^\alpha/K^\alpha}\right) \le e^{-x^\alpha/K^\alpha}\mathbb{E}\left[e^{|X|^\alpha/K^\alpha}\right] \le 2e^{-x^\alpha/K^\alpha}$$

(ii) ⇒ (iii). Note that for $s \ge 1$,

$$\Gamma(s) = \int_0^\infty x^{s-1}e^{-x}\,\mathrm{d}x = s^s\int_0^\infty t^{s-1}e^{-st}\,\mathrm{d}t \le s^s\int_0^\infty e^{(s-1)(t-1)}e^{-st}\,\mathrm{d}t \le s^s e^{1-s}. \tag{1.14}$$

By (1.5), for $p \geq 1$, we have

$$\begin{aligned}\mathbb{E}|X|^p &= \int_0^\infty p x^{p-1} \mathbb{P}(|X| \geq x)\,\mathrm{d}x \leq 2\int_0^\infty p x^{p-1} e^{-cx^\alpha}\,\mathrm{d}x \\ &= 2\int_0^\infty p\left(\frac{t}{c}\right)^{\frac{p-1}{\alpha}} e^{-t}\frac{(t/c)^{1/\alpha-1}}{c\alpha}\,\mathrm{d}t = \frac{2pc^{-p/\alpha}}{\alpha}\int_0^\infty t^{\frac{p}{\alpha}-1}e^{-t}\,\mathrm{d}t = 2c^{-p/\alpha}\Gamma\left(1+\frac{p}{\alpha}\right).\end{aligned}$$

By (1.14) and the inequality $(1+x) \leq e^x$, we have

$$\begin{aligned}(\mathbb{E}|X|^p)^{1/p} \leq c^{-1/\alpha}\left[2\Gamma\left(1+\frac{p}{\alpha}\right)\right]^{1/p} &\leq 2^{1/p}(ce)^{-1/\alpha}\left(1+\frac{p}{\alpha}\right)^{\frac{1}{p}+\frac{1}{\alpha}} \\ &\leq 2^{1/p}c^{-1/\alpha}\left(1+\frac{p}{\alpha}\right)^{\frac{1}{\alpha}} \leq 2c^{-1/\alpha}\left(1+\frac{1}{\alpha}\right)^{\frac{1}{\alpha}} p^{1/\alpha}.\end{aligned}$$

Then it suffices to choose $\omega = 4c^{-1/\alpha}$ in (iii).

(iii) $\Rightarrow$ (i). Note that $k! \geq (k/e)^k$. For $t \leq \omega^{-\alpha}/e$,

$$\mathbb{E}\left[e^{t|X|^\alpha}\right] = 1 + \sum_{k=1}^\infty \frac{t^k \mathbb{E}|X|^{k\alpha}}{k!} \leq 1 + \sum_{k=1}^\infty \frac{(\omega^\alpha k t)^k}{k!} = 1 + \sum_{k=1}^\infty (e\omega^\alpha t)^k = \frac{1}{1-e\omega^\alpha t}.$$

We take $K = (2e)^{1/\alpha}\omega$ in (i). Then we finishes the proof. □

Remark. According to our proof, if $\|X\|_{\psi_\alpha} < \infty$, we have

$$\mathbb{P}(|X| \geq x) \leq 2\exp\left(-\frac{x^\alpha}{\|X\|_{\psi_\alpha}^\alpha}\right), \quad x > 0,$$

and we can control the L^p norm of X by

$$\|X\|_{L^p} \leq \left[2\Gamma\left(1+\frac{p}{\alpha}\right)\right]^{1/p}\|X\|_{\psi_\alpha} \leq 4p^{1/\alpha}\|X\|_{\psi_\alpha}, \quad p \geq 1. \tag{1.15}$$

In particular, if X is sub-Gaussian and $\mathbb{E}X = 0$, according to the proof of Theorem 1.5, we have

$$\mathbb{E}\left[e^{tX}\right] \leq \exp\left(6t^2\|X\|_{\psi_2}^2\right), \quad t \in \mathbb{R}. \tag{1.16}$$

If X is sub-exponential and $\mathbb{E}X = 0$, we have $\mathbb{E}|X|^p \leq 2\Gamma(1+p)\|X\|_{\psi_1}^p$, and

$$\begin{aligned}\mathbb{E}\left[e^{tX}\right] &= 1 + \sum_{k=2}^\infty \frac{t^k\mathbb{E}[X^k]}{k!} \leq 1 + \sum_{k=2}^\infty 2t^k\|X\|_{\psi_1}^k \\ &= 1 + \frac{2t^2\|X\|_{\psi_1}^2}{1-t\|X\|_{\psi_1}} \leq \exp\left(3t^2\|X\|_{\psi_1}^2\right), \quad t \leq \frac{1}{3\|X\|_{\psi_1}}.\end{aligned} \tag{1.17}$$

Hence $X \sim \mathrm{SE}\big(6\|X\|_{\psi_1}^2, 3\|X\|_{\psi_1}\big)$.

We can check that centering does not destroy the ψ_α-norm.

Lemma 1.21 (Centering)**.** *There exists a constant $C_\alpha > 0$ depending on α such that*

$$\|X - \mathbb{E}X\|_{\psi_\alpha} \leq C_\alpha\|X\|_{\psi_\alpha}$$

for all random variables X with $\|X\|_{\psi_\alpha} < \infty$.

Proof. By the triangle inequality,

$$\|X - \mathbb{E}X\|_{\psi_\alpha} \le \|X\|_{\psi_\alpha} + \|\mathbb{E}X\|_{\psi_\alpha}.$$

For any constant $t \in \mathbb{R}$, one can easily verify that $\|t\|_{\psi_\alpha} = (\log 2)^{-1/\alpha} t$. Then by Jensen's inequality,

$$\|\mathbb{E}X\|_{\psi_\alpha} \le (\log 2)^{-1/\alpha} |\mathbb{E}X| \le (\log 2)^{-1/\alpha} \mathbb{E}|X| \le 2(\log 2)^{-1/\alpha} \|X\|_{\psi_\alpha}.$$

Then we can take $C_\alpha = 1 + 2(\log 2)^{-1/\alpha}$. □

Lemma 1.22 (Hölder)**.** *Let $p^{-1} + q^{-1} = 1$, and let X, Y be random variables such that $\|X\|_{\psi_p} < \infty$ and $\|Y\|_{\psi_q} < \infty$. Then XY is subexponential, and*

$$\|XY\|_{\psi_1} \le \|X\|_{\psi_p} \|Y\|_{\psi_q}.$$

In particular,

(i) $\|X^p\|_{\psi_1} = \|X\|_{\psi_p}^p$.

(ii) if X and Y are sub-Gaussian variables, then XY is subexponential, and $\|XY\|_{\psi_1} \le \|X\|_{\psi_2} \|Y\|_{\psi_2}$.

Proof. Using the Young's inequality $ab \le a^p/p + b^q/q$. Then

$$\mathbb{E}\left[\exp\left(\frac{|XY|}{\|X\|_{\psi_p}\|Y\|_{\psi_q}}\right)\right] \le \mathbb{E}\left[\exp\left(\frac{|X|^p}{p\|X\|_{\psi_p}^p} + \frac{|Y|^q}{q\|Y\|_{\psi_q}^q}\right)\right] \le \mathbb{E}\left[\frac{1}{p}\exp\left(\frac{|X|^p}{\|X\|_{\psi_p}^p}\right) + \frac{1}{q}\exp\left(\frac{|Y|^q}{\|Y\|_{\psi_q}^q}\right)\right] \le 2.$$

The case $\|X^p\|_{\psi_1} = \|X\|_{\psi_p}^p$ follows directly from definition. □

Remark. Generally, we have the following chain of inequalities:

$$\|X\|_{L^1} \le \|X\|_{L^p} \le \|X\|_{L^q} \le \cdots \lesssim \|X\|_{\psi_1} \lesssim \|X\|_{\psi_p} \lesssim \|X\|_{\psi_q} \lesssim \cdots \lesssim \|X\|_{L^\infty},$$

where $1 < p < q < \infty$.

2 Gaussian Concentration

2.1 Entropy and Sub-Gaussianity

Definition 2.1 (Entropy)**.** For a non-negative random variable Y, the *entropy* of Y is defined as

$$\mathrm{Ent}(Y) = \mathbb{E}[Y \log Y] - \mathbb{E}Y \log(\mathbb{E}Y).$$

For a random variable X, the following lemma has established the connection between the entropy of e^{tX} and sub-Gaussianity.

Lemma 2.2 (Herbst)**.** *Suppose that random variable X satisfies*

$$\mathrm{Ent}(e^{tX}) = \mathbb{E}[tXe^{tX}] - \mathbb{E}[e^{tX}] \log \mathbb{E}[e^{tX}] \leq \frac{t^2\sigma^2}{2}\mathbb{E}[e^{tX}], \quad \textit{for all } t \in \mathbb{R}. \tag{2.1}$$

Then X is σ^2-sub-Gaussian. Conversely, if X is sub-Gaussian with proxy $\sigma^2/4$, then it satisfies (2.1)*.*

Proof. (i) Let $\mu = \mathbb{E}X$, and define function $\varphi : \mathbb{R}\backslash\{0\} \to \mathbb{R}, t \mapsto \frac{1}{t}\log \mathbb{E}[e^{t(X-\mu)}]$, then

$$\frac{\mathrm{d}\varphi}{\mathrm{d}t}(t) = \frac{1}{t}\frac{\mathbb{E}[(X-\mu)e^{t(X-\mu)}]}{\mathbb{E}[e^{t(X-\mu)}]} - \frac{1}{t^2}\log \mathbb{E}[e^{t(X-\mu)}] = \frac{1}{t}\frac{\mathbb{E}[Xe^{tX}]}{\mathbb{E}[e^{tX}]} - \frac{1}{t^2}\log \mathbb{E}[e^{tX}] \leq \frac{\sigma^2}{2}.$$

We can complete φ on $\mathbb{R}$ by redefining $\varphi(0) = \lim_{t\to 0}\varphi(t) = 0$. Then $\varphi(t) - t\sigma^2/2$ is non-increasing on $\mathbb{R}$, and X is σ^2-sub-Gaussian:

$$\log \mathbb{E}[e^{tX}] - \frac{t^2\sigma^2}{2} = t\varphi(t) - \frac{t^2\sigma^2}{2} \leq t\varphi(0) = 0.$$

(ii) Suppose X is $\frac{1}{4}\sigma^2$-sub-Gaussian, and define $Z = e^{tX}/\mathbb{E}[e^{tX}]$. To prove (2.1), it suffices to show that

$$\mathbb{E}[Z \log Z] \leq \frac{t^2\sigma^2}{2}.$$

Suppose $Z \sim F$. Since Z is non-negative and $\mathbb{E}Z = 1$, we can define a new probability measure G such that $\mathrm{d}G(z) = z\,\mathrm{d}F(z)$. Then by Jensen's inequality, we have

$$\mathbb{E}[Z \log Z] = \int z \log z \mathrm{d}F(z) = \int \log z \mathrm{d}G(z) \leq \log\left(\int z \mathrm{d}G(z)\right) = \log \mathbb{E}[Z^2].$$

Furthermore, note that $\mathbb{E}[e^{t(X-\mu)}] \geq e^{\mathbb{E}[t(X-\mu)]} = 1$, we have $Z \leq e^{t(X-\mu)}$, and

$$\mathbb{E}[Z \log Z] \leq \log \mathbb{E}[Z^2] \leq \log \mathbb{E}[e^{2t(X-\mu)}] \leq \frac{(2t)^2\sigma^2}{8} = \frac{t^2\sigma^2}{2},$$

where the last equality follows from the sub-Gaussianity of X. Hence we conclude the proof. □

2.2 Lipschitz Function of Gaussian Variables

Before we proceed, we first introduce a logarithmic Sobolev inequality.

Lemma 2.3 (Gaussian log-Sobolev inequality)**.** *Let $\mathrm{d}\mu(x) = (2\pi)^{-\frac{n}{2}}e^{-\frac{1}{2}|x|^2}\mathrm{d}x$ be the standard Gaussian measure on $\mathbb{R}^n$. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a smooth function such that $f \geq 0$ and $f \in L^1(\mu)$. Then*

$$\int_{\mathbb{R}^n} f \log f \,\mathrm{d}\mu \leq \frac{1}{2}\int_{\mathbb{R}^n} \frac{|\nabla f|^2}{f}\,\mathrm{d}\mu + \|f\|_{L^1(\mu)} \log \|f\|_{L^1(\mu)}. \tag{2.2}$$

Proof. The proof is based on the semigroup theory for heat kernels. Given any $t > 0$, define P_t by

$$(P_t f)(x) = \frac{1}{(4\pi t)^{n/2}} \int_{\mathbb{R}^n} f(x) e^{-\frac{|x|^2}{4t}}\, dx, \quad f \in C_0^\infty(\mathbb{R}^n).$$

The generator for the semigroup $(P_t)_{t\geq 0}$ is the Laplacian operator Δ, and $P_t \Delta f = \Delta P_t f$. Furthermore,

$$\begin{aligned}
&\frac{\mathrm{d}}{\mathrm{d}s} P_s(P_{t-s} f \log P_{t-s} f) = P_s\left[\Delta(P_{t-s} f \log P_{t-s} f) - (1 + \log P_{t-s} f)\, \Delta P_{t-s} f\right] \\
&\quad = P_s\left[(\Delta P_{t-s} f) \log P_{t-s} f + 2\nabla P_{t-s} f \cdot \frac{\nabla P_{t-s} f}{P_{t-s} f} + P_{t-s} f\, \nabla \cdot \frac{\nabla P_{t-s} f}{P_{t-s} f} - (1 + \log P_{t-s} f)\, \Delta P_{t-s} f\right] \\
&\quad = P_s\left(\frac{|\nabla P_{t-s} f|^2}{P_{t-s} f}\right).
\end{aligned}$$

By Cauchy-Schwarz inequality,

$$|\nabla P_{t-s} f|^2 = |P_{t-s} \nabla f|^2 \leq (P_{t-s} |\nabla f|)^2 \leq P_{t-s}\left(\frac{|\nabla f|^2}{f}\right) P_{t-s} f$$

We use the fundamental theorem of calculus and the last two displays to obtain

$$\begin{aligned}
P_t(f \log f) - P_t f \log P_t f &= \int_0^t \frac{\mathrm{d}}{\mathrm{d}s} P_s(P_{t-s} f \log P_{t-s} f)\, \mathrm{d}s \\
&= \int_0^t P_s\left(\frac{|\nabla P_{t-s} f|^2}{P_{t-s} f}\right) \mathrm{d}s \\
&\leq \int_0^t P_s P_{t-s}\left(\frac{|\nabla f|^2}{f}\right) \mathrm{d}s = t P_t\left(\frac{|\nabla f|^2}{f}\right).
\end{aligned}$$

In particular, we take $t = \frac{1}{2}$ and evaluate both sides at $x = 0$:

$$\int_{\mathbb{R}^n} f \log f\, \mathrm{d}\mu - \int_{\mathbb{R}^n} f\, \mathrm{d}\mu \log\left(\int_{\mathbb{R}^n} f\, \mathrm{d}\mu\right) \leq \frac{1}{2} \int_{\mathbb{R}^n} \frac{|\nabla f|^2}{f}\, \mathrm{d}\mu.$$

Therefore

$$\int_{\mathbb{R}^n} f \log f\, \mathrm{d}\mu \leq \frac{1}{2} \int_{\mathbb{R}^n} \frac{|\nabla f|^2}{f}\, \mathrm{d}\mu + \|f\|_{L^1(\mu)} \log \|f\|_{L^1(\mu)}.$$

Thus we complete the proof. □

Remark. We can understand (2.2) from an information-theoretic perspective. Define $g = f/\|f\|_{L^1(\mu)}$ and $\mathrm{d}\nu = g\, \mathrm{d}\mu$, it can be verified that ν is also a probability measure on $\mathbb{R}^n$, and $g = \mathrm{d}\nu/\mathrm{d}\mu$ is the Radon-Nikodym derivative of ν with respect to μ. Moreover, (2.2) can be written as

$$\int g \log g\, \mathrm{d}\mu \leq \frac{1}{2} \int \frac{|\nabla g|^2}{g} \mathrm{d}\mu. \tag{2.3}$$

The LHS is the Kullback-Leibler divergence (or relative entropy) from μ to ν, and the RHS is half the relative Fisher information:

$$D_{\mathrm{KL}}(\nu \| \mu) := \int \log\left(\frac{\mathrm{d}\nu}{\mathrm{d}\mu}\right) \mathrm{d}\nu \leq \frac{1}{2} \int |\nabla \log g|^2\, \mathrm{d}\nu =: \frac{1}{2} \mathcal{I}(\nu \| \mu).$$

Therefore, this lemma gives an upper bound for the Kullback-Leibler divergence between ν and μ in terms of their relative Fisher information.

Theorem 2.4 (Gaussian concentration)**.** *Let $X \sim N(0, I_n)$, and let $f : \mathbb{R}^n \to \mathbb{R}$ be an L-Lipschitz continuous function. Then $f(X)$ is a sub-Gaussian variable with variance proxy L^2.*

Proof. Step I. Fix a smooth function $h \geq 0$ with $\|h\|_{L^1(\mu)} = \int |h| \mathrm{d}\mu > 0$. By Theorem 2.3,

$$\int h \log h \, \mathrm{d}\mu - \|h\|_{L^1(\mu)} \log \|h\|_{L^1(\mu)} \leq \frac{1}{2} \int \frac{|\nabla h|^2}{h} \mathrm{d}\mu. \tag{2.4}$$

Suppose $f \in C^\infty(\mathbb{R}^n)$ and f is L-Lipschitz continuous. Fix $t \in \mathbb{R}$ and set $h = e^{tf}$. By (2.4), we have

$$\mathrm{Ent}\left(e^{tf(X)}\right) \leq \frac{t^2}{2} \mathbb{E}\left[e^{tf(X)} |\nabla f(X)|^2\right] \leq \frac{t^2 L^2}{2} \mathbb{E}\left[e^{tf(X)}\right],$$

where the last equality holds because f is L-Lipschitz continuous. By Lemma 2.2, $f(X)$ is L^2-sub-Gaussian.

Step II. Now it remains to show that the conclusion holds for all L-Lipschitz f. (f is not necessarily differentiable.) Choose a non-negative $\psi \in C_c^\infty(\mathbb{R}^n)$ such that $\mathrm{supp}(\psi) \subseteq \{x \in \mathbb{R}^n : \|x\| \leq 1\}$ and $\int \psi(x) \mathrm{d}x = 1$, and define $\psi_\epsilon(x) := \frac{1}{\epsilon} \psi\left(\frac{x}{\epsilon}\right)$ for $\epsilon > 0$. Then $\int \psi_\epsilon(x) \mathrm{d}x = 1$.

Fix $\epsilon > 0$, and define $f_\epsilon = \psi_\epsilon * f : x \mapsto \int \psi_\epsilon(x - y) f(y) \mathrm{d}y$. Then $f_\epsilon \in C^\infty(\mathbb{R}^n)$, and f_ϵ is L-Lipschitz:

$$\begin{aligned} |f_\epsilon(x) - f_\epsilon(x')| &\leq \int \psi_\epsilon(y) \big| f(x - y) - f(x' - y) \big| \mathrm{d}y \\ &\leq \int \psi_\epsilon(y) L \|x - x'\|_2 \mathrm{d}y \leq L \|x - x'\|_2. \end{aligned}$$

Moreover, f_ϵ converges uniformly to f as $\epsilon \to 0$:

$$\begin{aligned} \|f_\epsilon - f\|_\infty &= \sup_{x \in \mathbb{R}^n} |f_\epsilon(x) - f(x)| = \sup_{x \in \mathbb{R}^n} \left| \int \psi_\epsilon(x - y) \left(f(y) - f(x)\right) \mathrm{d}y \right| \\ &\leq \sup_{x \in \mathbb{R}^n} \int_{\|y - x\|_2 \leq \epsilon} \psi_\epsilon(x - y) L \, \|y - x\|_2 \, \mathrm{d}y \leq \epsilon L \int_{\|y\|_2 \leq \epsilon} \psi_\epsilon(-y) \mathrm{d}y = \epsilon L. \end{aligned}$$

Step III. Fix $t \in \mathbb{R}$. For any $\epsilon > 0$ and $x \in \mathbb{R}^n$, we have $e^{tf(x)} \leq e^{tf_\epsilon(x) + |t|\epsilon L}$. Moreover, $f_\epsilon \in C_c^\infty(\mathbb{R}^n)$ and f_ϵ is continuous, then $f_\epsilon(X)$ is L^2-sub-Gaussian. Therefore

$$\mathbb{E}\left[e^{tf(X)}\right] \leq \inf_{\epsilon > 0} \mathbb{E}\left[e^{tf_\epsilon(X)}\right] e^{|t|\epsilon L} \leq \inf_{\epsilon > 0} \exp\left(\frac{t^2 L^2}{2} + |t|\epsilon L\right) = \exp\left(\frac{t^2 L^2}{2}\right),$$

which concludes the proof. □

3 Tail Bounds

3.1 Chernoff-type Bounds and Hoeffding Inequalities

The Chernoff bound gives an important tail bound for the deviation of mean from the true expectation in the sub-Gaussian case.

Theorem 3.1 (Hoeffding)**.** *Let $X_1, \cdots, X_n$ be independent sub-Gaussian variables with variance proxies $\sigma_1^2, \cdots, \sigma_n^2$. Then*

$$\mathbb{P}\left(\sum_{i=1}^{n}(X_i - \mathbb{E}X_i) \geq \epsilon\right) \leq \exp\left(-\frac{\epsilon^2}{2\sum_{i=1}^{n}\sigma_i^2}\right). \tag{3.1}$$

Proof. It can be easily verified that $\sum_{i=1}^{n}(X_i - \mathbb{E}X_i)$ is a sub-Gaussian variable with mean 0 and variance proxy $\sum_{i=1}^{n}\sigma_i^2$. Then (3.1) immediately follows from (1.2) in Theorem 1.5. □

Combining Lemma 1.9 and Theorem 3.1 gives the following Hoeffding's inequality:

Theorem 3.2 (Hoeffding inequality for bounded variables)**.** *Let $X_1, \cdots, X_n$ be independent random variables such that $\mathbb{P}(X_i \in [a_i, b_i]) = 1$ for $i = 1, \cdots, n$. Then*

$$\mathbb{P}\left(\sum_{i=1}^{n}(X_i - \mathbb{E}X_i) \geq \epsilon\right) \leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^{n}(b_i - a_i)^2}\right).$$

Also, we can combine the estimate (1.16) and Theorem 3.1 to obtain the following bound.

Theorem 3.3 (Hoeffding)**.** *Let $X_1, \cdots, X_n$ be independent zero-mean sub-Gaussian variables. Then*

$$\mathbb{P}\left(\sum_{i=1}^{n} X_i \geq \epsilon\right) \leq \exp\left(-\frac{\epsilon^2}{24\sum_{i=1}^{n}\|X_i\|_{\psi_2}^2}\right). \tag{3.2}$$

The Hoeffding inequalities are extremely important concentration in statistical learning theory. To move forward, we can relax the independence assumption and extend the result to martingale difference sequences.

Theorem 3.4 (Azuma-Hoeffding)**.** *Let $(X_n)_{n=1}^{\infty}$ be a martingale difference sequence with respect to the filtration $(\mathscr{F}_n)_{n=0}^{\infty}$, i.e. X_n is $\mathscr{F}_n$-measurable and $\mathbb{E}[X_n|\mathscr{F}_{n-1}] = 0$ for $n \in \mathbb{N}$. If every X_n is conditionally sub-Gaussian with proxy σ_n^2, i.e.*

$$\mathbb{E}\left[e^{tX_n}|\mathscr{F}_{n-1}\right] \leq \exp\left(\frac{t^2\sigma_n^2}{2}\right) \quad a.s.,$$

then for every $n \in \mathbb{N}$,

$$\mathbb{P}\left(\sum_{i=1}^{n} X_i \geq \epsilon\right) \leq \exp\left(-\frac{\epsilon^2}{2\sum_{i=1}^{n}\sigma_i^2}\right). \tag{3.3}$$

In particular, if there exist predictable sequences $(A_n)_{n=1}^{\infty}$ and $(B_n)_{n=1}^{\infty}$ such that

$$A_n \leq X_n \leq B_n \quad \text{and} \quad B_n - A_n \leq c_n \quad a.s. \ \text{for } n = 1, 2, \cdots,$$

then for every $n \in \mathbb{N}$,

$$\mathbb{P}\left(\sum_{i=1}^{n} X_i \geq \epsilon\right) \leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^{n}c_i^2}\right). \tag{3.4}$$

Proof. For every $t > 0$, we have the Chernoff bound

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq \epsilon\right) \leq e^{-t\epsilon}\,\mathbb{E}\left[\exp\left(t\sum_{i=1}^n X_i\right)\right].$$

To exploit the martingale property, we condition on the filtration:

$$\mathbb{E}\left[\exp\left(t\sum_{i=1}^n X_i\right)\right] = \mathbb{E}\left[\mathbb{E}\left[\exp\left(t\sum_{i=1}^n X_i\right)\Big|\mathscr{F}_{n-1}\right]\right] = \mathbb{E}\left[\exp\left(t\sum_{i=1}^{n-1} X_i\right)\mathbb{E}\left[e^{tX_n}|\mathscr{F}_{n-1}\right]\right].$$

By the conditional sub-Gaussianity and induction, we have

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq \epsilon\right) \leq \exp\left(-t\epsilon + \frac{t^2}{2}\sum_{i=1}^n \sigma_i^2\right).$$

Setting $t = \epsilon/\sum_{i=1}^n \sigma_i^2$, we have

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq \epsilon\right) \leq \exp\left(-\frac{\epsilon^2}{2\sum_{i=1}^n \sigma_i^2}\right).$$

In particular, if X_n is a.s. bounded by $[A_n, B_n]$, where A_n, B_n are $\mathscr{F}_{n-1}$-measurable and $B_n - A_n \leq c_n$, then by Hoeffding's lemma,

$$\mathbb{E}\left[e^{tX_n}|\mathscr{F}_{n-1}\right] \leq \exp\left(\frac{t^2 c_n^2}{8}\right).$$

And the result (3.4) for marginales with bounded difference follows. □

Remark. In fact, we can enhance the results (3.3) and (3.4) by Doob's maximal inequality for submartingales. Since $x \mapsto e^{tx}$ is convex for every $t \in \mathbb{R}$, the sequence $\exp(t\sum_{i=1}^n X_i),\ n = 1, 2, \cdots$ is a submartingale. Then

$$\mathbb{P}\left(\sup_{1\leq k\leq n}\sum_{i=1}^k X_i \geq \epsilon\right) = \mathbb{P}\left(\sup_{1\leq k\leq n}\exp\left(t\sum_{i=1}^k X_i\right) \geq e^{t\epsilon}\right) \leq e^{-t\epsilon}\,\mathbb{E}\left[\exp\left(t\sum_{i=1}^n X_i\right)\right].$$

Hence (3.3) becomes

$$\mathbb{P}\left(\sup_{1\leq k\leq n}\sum_{i=1}^k X_i \geq \epsilon\right) \leq \exp\left(-\frac{\epsilon^2}{2\sum_{i=1}^n \sigma_i^2}\right),$$

and (3.4) becomes

$$\mathbb{P}\left(\sup_{1\leq k\leq n}\sum_{i=1}^k X_i \geq \epsilon\right) \leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n c_i^2}\right). \tag{3.5}$$

The power of the Azuma-Hoeffding inequality is that it produces tail inequalities for quantities other than sums of independent random variables.

Theorem 3.5 (McDiarmid's inequality)**.** *A function $f : \mathcal{X}_1 \times \cdots \times \mathcal{X}_n \to \mathbb{R}$ satisfies the bound difference property if substituting the value of the i-th coordinate x_i changes the value of f at most c_i. More formally, if there exists $c_1, \cdots, c_n \geq 0$ such that for all $x = (x_1, \cdots, x_n) \in \mathcal{X}_1 \times \cdots \times \mathcal{X}_n$,*

$$\sup_{y_i\in\mathcal{X}_i} f(x_1, \cdots, x_{i-1}, y_i, \cdots, x_n) - \inf_{y_i'\in\mathcal{X}_i} f(x_1, \cdots, x_{i-1}, y_i', \cdots, x_n) \leq c_i, \quad i = 1, 2, \cdots, n.$$

Let $X_1, \cdots, X_n$ be independent random variables valued in $\mathcal{X}_1, \cdots, \mathcal{X}_n$, respectively. Then

$$\mathbb{P}\left(f(X_1, \cdots, X_n) - \mathbb{E}\left[f(X_1, \cdots, X_n)\right] \geq \epsilon\right) \leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n c_i^2}\right),$$
$$\mathbb{P}\left(f(X_1, \cdots, X_n) - \mathbb{E}\left[f(X_1, \cdots, X_n)\right] \leq -\epsilon\right) \leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n c_i^2}\right).$$

As a result,

$$\mathbb{P}\left(\left|f(X_1, \cdots, X_n) - \mathbb{E}\left[f(X_1, \cdots, X_n)\right]\right| \geq \epsilon\right) \leq 2\exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n c_i^2}\right).$$

Proof. We fix the canonical filtration $\mathscr{F}_i = \sigma(X_1, \cdots, X_i)$, $i = 0, 1, \cdots, n$, and consider Doob's martingale

$$Z_i = \mathbb{E}\left[f(X)|\mathscr{F}_i\right], \quad i = 0, 1, \cdots, n.$$

We write $X = (X_1, \cdots, X_n)$. For each $1 \leq i \leq n$, define

$$A_i = \inf_{x \in \mathcal{X}_i} \mathbb{E}\left[f(X_1, \cdots, X_{i-1}, x, X_{i+1}, \cdots, X_n)|\mathscr{F}_{i-1}\right] - E\left[f(X)|\mathscr{F}_{i-1}\right],$$
$$B_i = \sup_{x \in \mathcal{X}_i} \mathbb{E}\left[f(X_1, \cdots, X_{i-1}, x, X_{i+1}, \cdots, X_n)|\mathscr{F}_{i-1}\right] - E\left[f(X)|\mathscr{F}_{i-1}\right].$$

Then both $(A_i)_{i=1}^n$ and $(B_i)_{i=1}^n$ are predictable, $A_i \leq Z_i - Z_{i-1} \leq B_i$ a.s., and

$$B_i - A_i = \sup_{x, y \in \mathcal{X}_i} \mathbb{E}\left[f(X_1, \cdots, X_{i-1}, x, X_{i+1}, \cdots, X_n) - f(X_1, \cdots, X_{i-1}, y, X_{i+1}, \cdots, X_n)|\mathscr{F}_{i-1}\right] \leq c_i.$$

An application of the Azuma-Hoeffding inequality (3.4) gives

$$\begin{aligned}\mathbb{P}\left(f(X_1, \cdots, X_n) - \mathbb{E}\left[f(X_1, \cdots, X_n)\right] \geq \epsilon\right) &= \mathbb{P}(Z_n - Z_0 \geq \epsilon) \\ &= \mathbb{P}\left(\sum_{i=1}^n (Z_i - Z_{i-1}) \geq \epsilon\right) \leq \exp\left(-\frac{2\epsilon^2}{\sum_{i=1}^n c_i^2}\right).\end{aligned}$$

Then we complete the proof. □

Remark. Given spaces $\mathcal{X}_1, \cdots, \mathcal{X}_n$, we define the *Hamming metric* on the product space $\mathcal{X}_1 \times \cdots \times \mathcal{X}_n$ by

$$d_{\mathbf{1}}(x, y) = \sum_{i=1}^n \mathbb{1}_{\{x_i \neq y_i\}}$$

Furthermore, given weights $\alpha_1, \cdots, \alpha_n \geq 0$, we further define the *weighted Hamming metric* on the product space $\mathcal{X}_1 \times \cdots \times \mathcal{X}_n$ by

$$d_\alpha(x, y) = \sum_{i=1}^n \alpha_i \mathbb{1}_{\{x_i \neq y_i\}}. \tag{3.6}$$

Then if f is 1-Lipshictz under the metric d_α, it satisfies the bounded difference property

$$\sup_{y_i \in \mathcal{X}_i} f(x_1, \cdots, x_{i-1}, y_i, \cdots, x_n) - \inf_{y_i' \in \mathcal{X}_i} f(x_1, \cdots, x_{i-1}, y_i', \cdots, x_n) \leq \alpha_i, \quad i = 1, 2, \cdots, n,$$

and we can apply McMiarmid's inequality.

3.2 Sub-Gaussian Khintchine Inequality

For independent L^2 random variables $X_1, \cdots, X_n$, one have the Pythagorean theorem:

$$\left\| \sum_{i=1}^n X_i \right\|_{L^2} = \sum_{i=1}^n \|X_i\|_{L^2}.$$

For independent sub-Gaussian random variables $X_1, \cdots, X_n$, we have a relaxed version of the above theorem for the sub-Gaussian norm.

Proposition 3.6. *Let $X_1, \cdots, X_n$ be independent sub-Gaussian variables. Then*

$$\left\| \sum_{i=1}^n X_i \right\|_{\psi_2}^2 \leq 36 \sum_{i=1}^n \|X_i\|_{\psi_2}^2$$

Proof. By independence of $X_1, \cdots, X_n$ and (1.16),

$$\mathbb{E}\left[e^{t(X_1+\cdots+X_n)}\right] = \prod_{i=1}^n \mathbb{E}\left[e^{tX_i}\right] \leq \prod_{i=1}^n e^{6t^2\|X_i\|_{\psi_2}^2} = e^{6Kt^2}.$$

where $K^2 = \sum_{i=1}^n \|X_i\|_{\psi_2}^2$. Using the direction (i) $\Rightarrow$ (iii) in Theorem 1.5, we have

$$\mathbb{E}\left[\exp\left(\frac{(X_1+\cdots+X_n)^2}{36K^2}\right)\right] \leq 2.$$

Therefore $\| \sum_{i=1}^n X_i\|_{\psi_2}^2 \leq 36K^2 = 36 \sum_{i=1}^n \|X_i\|_{\psi_2}^2$. □

We have the following estimate for the L^p norm of sum of sub-Gaussian random variables.

Theorem 3.7 (Khintchine's inequality)**.** *Let $X_1, \cdots, X_n$ be independent zero-mean sub-Gaussian variables, and $p \geq 2$. Then for any $a = (a_1, \cdots, a_n) \in \mathbb{R}^n$, it holds*

$$\|a\|_2 \min_{1\leq i\leq n} \|X_i\|_{L^2} \leq \left\| \sum_{i=1}^n a_i X_i \right\|_{L^p} \leq 12\sqrt{p}\,\|a\|_2 \max_{1\leq i\leq n} \|X_i\|_{\psi_2}$$

Proof. For the lower bound, note that

$$\left\| \sum_{i=1}^n a_i X_i \right\|_{L^p} \geq \left\| \sum_{i=1}^n a_i X_i \right\|_{L^2} = \left[\sum_{i=1}^n |a_i|^2 \mathbb{E}|X_i|^2\right]^{1/2} \geq \|a\|_2 \min_{1\leq i\leq n} \|X_i\|_{L^2}.$$

For the upper bound, we use Proposition 3.6:

$$\left\| \sum_{i=1}^n a_i X_i \right\|_{\psi_2} \leq 6 \left[\sum_{i=1}^n |a_i|^2 \|X_i\|_{\psi_2}^2\right]^{1/2} \leq 6\|a\|_2 \max_{1\leq i\leq n} \|X_i\|_{\psi_2}.$$

By (1.15),

$$\left\| \sum_{i=1}^n a_i X_i \right\|_{L^p} \leq 2\sqrt{p} \left\| \sum_{i=1}^n a_i X_i \right\|_{\psi_2} \leq 12\sqrt{p}\,\|a\|_2 \max_{1\leq i\leq n} \|X_i\|_{\psi_2}.$$

Thus we finish the proof. □

Remark. If $X_1, \cdots, X_n$ are i.i.d. Rademacher variables, we can get a sharper upper bound. Note that

$$\mathbb{E}\left[e^{t(a_1X_1+\cdots+a_nX_n)}\right] \le \exp\left(\frac{t^2\|a\|_2^2}{2}\right), \quad t \in \mathbb{R}.$$

Using the direction (i) $\Rightarrow$ (iii) in Theorem 1.5, we have

$$\mathbb{E}\left[\exp\left(\frac{|a_1X_1+\cdots+a_nX_n|^2}{6\|a\|_2^2}\right)\right] \le 2.$$

Hence by (1.15) we have

$$\left\|\sum_{i=1}^n a_iX_i\right\|_{L^p} \le 2\sqrt{p}\left\|\sum_{i=1}^n a_iX_i\right\|_{\psi_2} \le 2\sqrt{6p}\,\|a\|_2.$$

More generally, if $X_1, \cdots, X_n$ are independent zero-mean sub-Gaussian variables with proxies $\sigma_1^2, \cdots, \sigma_n^2$, then

$$\left\|\sum_{i=1}^n a_iX_i\right\|_{L^p} \le 2\sqrt{p}\left\|\sum_{i=1}^n a_iX_i\right\|_{\psi_2} \le 2\sqrt{6p\max_{1\le i\le n}\sigma_i^2}\,\|a\|_2.$$

3.3 Maximum of Sub-Gaussian Variables

Suppose we have n centered independent sub-Gaussian variables with variance proxy σ^2. A natural tail bound for the maximum can be attained from the fact that $\{\max_{1\le i\le n} X_i \ge \epsilon\} = \bigcup_{i=1}^n \{X_i \ge \epsilon\}$:

$$\mathbb{P}\left(\max_{1\le i\le n} X_i \ge \epsilon\right) \le n\exp\left(-\frac{\epsilon^2}{2\sigma^2}\right). \tag{3.7}$$

We can also bound the expected value as follows.

Theorem 3.8. *Let $X_1, \cdots, X_n$ be independent σ^2-sub-Gaussian variables with mean zero. Then*

$$\mathbb{E}\left[\max_{1\le i\le n} X_i\right] \le \sigma\sqrt{2\log n}.$$

Proof. Fix $\epsilon > 0$. By Jensen's inequality, we have

$$\begin{aligned}\mathbb{E}\left[\max_{1\le i\le n} X_i\right] &\le \frac{1}{t}\log\mathbb{E}\left[\exp\left(t\max_{1\le i\le n} X_i\right)\right] \le \frac{1}{t}\log\mathbb{E}\left[\sum_{i=1}^n e^{tX_i}\right] \\ &\le \frac{1}{t}\log\left\{\sum_{i=1}^n \exp\left(\frac{t^2\sigma^2}{2}\right)\right\} = \frac{\log n}{t} + \frac{t\sigma^2}{2}.\end{aligned}$$

Then we conclude the proof by setting $t = \sqrt{2\log n}/\sigma$. □

An immediate corollary of this theorem is the Massart's finite class lemma.

Lemma 3.9 (Massart)**.** *Let $\mathcal{A}$ be a finite subset of $\mathbb{R}^n$ and $\epsilon_1, \cdots, \epsilon_n$ be independent Rademacher variables. Denote by $r_{\mathcal{A}} = \max_{a\in\mathcal{A}}\|a\|_2$ the radius of $\mathcal{A}$. Then we have*

$$\mathbb{E}\left[\max_{a\in\mathcal{A}}\frac{1}{n}\sum_{i=1}^m \epsilon_i a_i\right] \le \frac{r_{\mathcal{A}}\sqrt{2\log|\mathcal{A}|}}{n}.$$

We can generalize the preceding bound to non-independent sub-Gaussian variables.

3.4 Bernstein Inequalities

In this subsection, we discuss several kinds of Bernstein inequalities.

Theorem 3.10 (Bernstein inequality for bounded variables)**.** *Let $X_1, \cdots, X_n$ be independent random variables such that $|X_i - \mathbb{E}X_i| \le b$ a.s. for $i = 1, \cdots, n$. Then for each $\epsilon > 0$,*

$$\mathbb{P}\left(\sum_{i=1}^{n}(X_i - \mathbb{E}[X_i]) \ge \epsilon\right) \le \exp\left(-\frac{\epsilon^2}{2(\sum_{i=1}^{n}\mathbb{E}[X_i^2] + b\epsilon/3)}\right).$$

Proof. Without loss of generality we assume $\mathbb{E}X_i = 0$ for all i. Since $X_1, \cdots, X_n$ are indepndent, and following (1.12), we have

$$\mathbb{E}[e^{t(X_1+\cdots+X_n)}] \le \exp\left(\frac{t^2\sum_{i=1}^{n}\mathbb{E}[X_i^2]}{2(1-bt/3)}\right).$$

Taking $t = \epsilon/(\sum_{i=1}^{n}\mathbb{E}[X_i^2] + b\epsilon/3)$, we have

$$\mathbb{P}\left(\sum_{i=1}^{n}X_i \ge \epsilon\right) \le \exp\left(-\frac{\epsilon^2}{2(\sum_{i=1}^{n}\mathbb{E}[X_i^2] + b\epsilon/3)}\right)$$

This is the desired result. □

Theorem 3.11 (Bernstein inequality for sub-exponential variables)**.** *Let $X_i \sim \mathrm{SE}(\nu_i^2, \alpha_i)$, $i = 1, \cdots, n$ be independent sub-exponential random variables. Then*

$$\mathbb{P}\left(\sum_{i=1}^{n}(X_i - \mathbb{E}[X_i]) \ge \epsilon\right) \le \exp\left(-\frac{1}{2}\min\left\{\frac{\epsilon^2}{\sum_{i=1}^{n}\nu_i^2}, \frac{\epsilon}{\max_{1\le i\le n}\alpha_i}\right\}\right).$$

Proof. This is an immediate corollary of Proposition 1.13 and Theorem 1.11. □

Theorem 3.12 (Bernstein inequality for sub-exponential variables)**.** *Let $X_1, X_2, \cdots, X_n$ be independent, zero-mean sub-exponential random variables. Then*

$$\mathbb{P}\left(\sum_{i=1}^{n}X_i \ge \epsilon\right) \le \exp\left(-\frac{1}{6}\min\left\{\frac{\epsilon^2}{2\sum_{i=1}^{n}\|X_i\|_{\psi_1}^2}, \frac{\epsilon}{\max_{1\le i\le n}\|X_i\|_{\psi_1}}\right\}\right).$$

Proof. This is an immediate corollary of (1.17), Proposition 1.13 and Theorem 1.11. □

3.5 Talagrand's Inequality

In this subsection, we consider the concentration of measures around a set. The main tool we use is Talagrand's convex distance, which evaluates the distance between a point and a set.

Definition 3.13 (Talagrand's convex distance)**.** Let $\mathcal{X} = \mathcal{X}_1 \times \cdots \times \mathcal{X}_n$ be a product space. Recall that the α-weighted Hamming distance d_α is given by (3.6). If $A \subset \mathcal{X}$ and $x \in \mathcal{X}$, the *convex distance* from x to A is

$$d_{\mathrm{C}}(x, A) = \sup_{\alpha\in\mathbb{R}_+^n:\|\alpha\|_2=1} d_\alpha(x, A),$$

where $d_\alpha(x, A) = \inf_{y\in A} d_\alpha(x, y)$.

We have the following equivalent definition of Talagrand's convex distance, which is a little opaque.

Proposition 3.14 (Equivalent definition of Talagrand's convex distance)**.** *Let $\mathcal{X} = \mathcal{X}_1 \times \cdots \times \mathcal{X}_n$ be a product space. If $A \subset \mathcal{X}$ and $x \in \mathcal{X}$, define*

$$U_A(x) = \{u \in \{0,1\}^n : \textit{there exists } y \in A \textit{ such that } x_i = y_i \textit{ for all } u_i = 0\}.$$

Let $V_A(x) \subset \mathbb{R}^n$ be the convex hull of $U_A(x)$. Then

$$d_{\mathrm{C}}(x, A) = \min_{v \in V_A(x)} \|v\|_2,$$

Remark. We note that $U_A(x)$ is always nonempty because it contains the all-one vector. Indeed, one can explain $U_A(x)$ as a set of travel plans: if $u \in U_A(x)$, one can reach A from x by varying only the coordinates $i \in \{1, \cdots, n\}$ for which $u_i = 1$.

Proof. Fix $A \subset \mathcal{X}$ and $x \in \mathcal{X}$. We let $U'_A(x) = \{u = (\mathbb{1}_{\{x_i \neq y_i\}})_{i=1}^n : y \in A\} \subset \{0,1\}^n$. Then

$$U'_A(x) \subset U_A(x) = \left\{u \in \{0,1\}^n : \text{there exists } y \in A \text{ such that } u_i \geq \mathbb{1}_{\{x_i \neq y_i\}} \text{ for all } i\right\}.$$

We take $\alpha \in \mathbb{R}^n_+$ with $\|\alpha\|_2 = 1$. Then

$$d_\alpha(x, A) = \inf_{y \in A} \sum_{i=1}^n \alpha_i \mathbb{1}_{\{x_i \neq y_i\}} = \inf_{u \in U'_A(x)} \langle \alpha, u \rangle = \inf_{u \in U_A(x)} \langle \alpha, u \rangle = \inf_{v \in V_A(x)} \langle \alpha, v \rangle,$$

where the last inequality holds because the linear functional $\langle \alpha, \cdot \rangle$ attains its infimum on a compact convex set $V_A(x)$ on its extreme points $U_A(x)$. Then

$$d_{\mathrm{C}}(x, A) = \sup_{\alpha \in \mathbb{R}^n_+ : \|\alpha\|_2 = 1} d_\alpha(x, A) = \sup_{\alpha \in \mathbb{R}^n_+ : \|\alpha\|_2 = 1} \inf_{v \in V_A(x)} \langle \alpha, v \rangle \leq \min_{v \in V_A(x)} \|v\|_2.$$

To derive the reverse inequality, we let $w \in V_A(x)$ be such that $\|w\|_2 = \min_{v \in V_A(x)} \|v\|_2$. If $\|w\|_2 = 0$ we are done. Otherwise, let $\alpha = w/\|w\|_2$. Then for all $v \in V_A(x)$ and $t \in (0,1)$, by convexity we have $w + t(v - w) \in V_A$, and

$$\|w\|_2 \leq \|w + t(v - w)\|_2^2 = \|w\|_2^2 + 2t\langle w, v - w \rangle + t^2 \|v - w\|_2^2$$

Hence $2\langle w, v - w \rangle + t\|v - w\|_2^2 \geq 0$. Letting $t \downarrow 0$, we have $\langle w, v - w \rangle \geq 0$, and it follows that $\langle \alpha, v \rangle \geq \|w\|_2$. Therefore

$$\mathrm{d}_{\mathrm{C}}(x, A) \geq d_\alpha(x, A) = \inf_{v \in V_A(x)} \langle \alpha, v \rangle \geq \|w\|_2 = \min_{v \in V_A(x)} \|v\|_2.$$

Then we finish the proof. □

Now we introduce the concentration result.

Theorem 3.15 (Talagrand's inequality)**.** *Let $\mathcal{X} = \mathcal{X}_1 \times \cdots \times \mathcal{X}_n$, and let $\mu = \mu_1 \otimes \cdots \otimes \mu_n$ be a product measure on $\mathcal{X}$. For all nonempty measurable subsets $A \subset \mathcal{X}$,*

$$\int_{\mathcal{X}} \exp\left(\frac{d_{\mathrm{C}}(x, A)^2}{4}\right) \mathrm{d}\mu(x) \leq \frac{1}{\mu(A)}.$$

Consequently,

$$\mu\left\{x \in \mathcal{X} : d_{\mathrm{C}}(x, A) \geq \epsilon\right\} \leq \frac{1}{\mu(A)} e^{-\epsilon^2/4}.$$

Proof. We proceed by induction. For $n = 1$, the convex distance $d_C(x, A) = \mathbb{1}_{\{x \notin A\}}$. Then

$$\int_{\mathcal{X}} \exp\left(\frac{d_{\mathrm{C}}(x, A)}{4}\right) \mathrm{d}\mu(x) = \mu(A) + e^{1/4}(1 - \mu(A)) \leq 1 \leq \frac{1}{\mu(A)}.$$

Next, we assume the result holds for dimension n. We fix a product space $\mathcal{Y} = \mathcal{Y}_1 \times \cdots \times \mathcal{Y}_n$ equipped with product measure $\nu = \mu_1 \otimes \cdots \otimes \mu_n$ and a new probability space $(\mathcal{Z}, \mu_{n+1})$. Let $\mathcal{X} = \mathcal{Y} \times \mathcal{Z}$, and $\mu = \nu \otimes \mu_{n+1}$. We fix nonempty $A \subset \mathcal{X}$ and $x = (y, z) \in \mathcal{X}$, where $y = (y_1, \cdots, y_n) \in \mathcal{Y}$ and $z \in \mathcal{Z}$.

Step I. Let $A_z = \{y \in \mathcal{Y} : (y, z) \in A\}$ be the z-slice of A, and $B = \bigcup_{z \in \mathcal{Z}} A_z$ be the projection of A onto $\mathcal{Y}$. We note the following relations among $U_A(x)$, $U_{A_z}(y)$ and $U_B(y)$:

- If $u \in U_{A_z}(y)$, then there exists $(y', z) \in A$ such that $y'_i = y_i$ for all $u_i = 0$, and $(u, 0) \in U_A(x)$;
- If $u \in U_B(y)$, then there exists $y' \in B$ such that $y'_i = y_i$ for all $u_i = 0$, and $(u, 1) \in U_A(x)$.

Now if $\xi \in V_{A_x}(y)$ and $\eta \in V_B(y)$, then both $(\xi, 0)$ and $(\zeta, 1)$ are in $V_A(x)$. By convexity of $V_A(x)$, the point $t(\xi, 0) + (1 - t)(\zeta, 1) \in V_A(x)$ for all $t \in [0, 1]$, and

$$\begin{aligned} d_{\mathrm{C}}(x, A)^2 &= \inf_{v \in V_A(x)} \|v\|_2^2 \leq \|(t\xi + (1-t)\zeta, 1-t)\|_2^2 \\ &\leq \|t\xi + (1-t)\zeta\|_2^2 + (1-t)^2 \leq t\|\xi\|_2^2 + (1-t)\|\eta\|^2 + (1-t)^2. \end{aligned}$$

Since $\xi \in V_{A_x}(y)$ and $\eta \in V_B(y)$ are arbitrary, we have

$$d_{\mathrm{C}}(x, A)^2 \leq t d_{\mathrm{C}}(y, A_x)^2 + (1-t) d_{\mathrm{C}}(y, B)^2 + (1-t)^2, \quad t \in [0, 1]. \tag{3.8}$$

Note that if $A_x = \emptyset$, we have $d(y, A_x) = \infty$, and the above estimate is saimply $d_{\mathrm{C}}(x, A) \leq d_{\mathrm{C}}(y, B)^2 + 1$.

Lemma 3.16. *For all $x \geq 1$, we have*

$$\inf_{t \in [0,1]} x^t e^{(1-t)^2/4} \leq 2 - \frac{1}{x}.$$

Proof. We take $t = 1 - 2\log x$ for $x \leq \sqrt{e}$, and $t = 0$ for $x \geq \sqrt{e}$. Then

$$e^{(1-t)^2/4} x^t \leq \begin{cases} e^{-(\log x)^2 - \log x}, & x \leq \sqrt{e}, \\ e^{1/4}, & x \geq \sqrt{e}. \end{cases}$$

If $x > \sqrt{e}$, we have $e^{1/4} \leq 2 - e^{-1/2} \leq 1 - x^{-1}$, and the result holds. If $x \leq \sqrt{e}$, then

$$x^t e^{(1-t)^2/4} + \frac{1}{x} \leq e^{-(\log x)^2 - \log x} + \frac{1}{x} = \frac{1}{x}\left(e^{-(\log x)^2} + 1\right) \leq 2.$$

Thus we complete the proof of the lemma. □

Step II. By (3.8) and the induction hypothesis,

$$\begin{aligned} \int_{\mathcal{Y}} \exp\left(\frac{d_C((y,z), A)^2}{4}\right) \mathrm{d}\nu(y) &\leq e^{(1-t)^2/4} \int_{\mathcal{Y}} \exp\left(\frac{t d_{\mathrm{C}}(y, A_x)^2}{4}\right) \exp\left(\frac{(1-t) d_{\mathrm{C}}(y, B)^2}{4}\right) \mathrm{d}\nu(y) \\ &\leq e^{(1-t)^2/4} \left(\int_{\mathcal{Y}} \exp\left(\frac{d_{\mathrm{C}}(y, A_x)^2}{4}\right) \mathrm{d}\nu(y)\right)^t \left(\int_{\mathcal{Y}} \exp\left(\frac{d_{\mathrm{C}}(y, B)^2}{4}\right) \mathrm{d}\nu(y)\right)^{1-t} \\ &\leq e^{(1-t)^2/4} \left(\frac{1}{\nu(A_z)}\right)^t \left(\frac{1}{\nu(B)}\right)^{1-t} = \frac{1}{\nu(B)} e^{(1-t)^2/4} \left(\frac{\nu(B)}{\nu(A_z)}\right)^t. \end{aligned}$$

Now we optimize the above bound by taking

$$t = \max\left\{0, 1 - 2\log\frac{\nu(B)}{\nu(A_z)}\right\}.$$

Then

$$\int_{\mathcal{Y}} \exp\left(\frac{d_{\mathrm{C}}((y,z),A)^2}{4}\right) \mathrm{d}\nu(y) \le \frac{1}{\nu(B)}\left(2 - \frac{\nu(A_z)}{\nu(B)}\right),$$

and

$$\int_{\mathcal{Z}}\int_{\mathcal{Y}} \exp\left(\frac{d_{\mathrm{C}}((y,z),A)^2}{4}\right) \mathrm{d}\nu(y)\,\mathrm{d}\mu_{n+1}(z) \le \frac{1}{\nu(B)}\left(2 - \frac{(\nu\otimes\mu_{n+1})(A)}{\nu(B)}\right) = \frac{1}{\nu(B)}\left(2 - \frac{\mu(A)}{\nu(B)}\right).$$

Note that $\frac{\mu(A)}{\nu(B)}\left(2 - \frac{\mu(A)}{\nu(B)}\right) \le 1$. Then we finish the induction step and conclude the proof. □

Now we see an application of Talagrand's inequality. Before we proceed, we first introduce a technical result concerning the relation between convex distance and Euclidean distance.

Lemma 3.17 (Convex distance controls Euclidean distance)**.** *Let $A \subset [0,1]^n$ be a convex set, and $x \in [0,1]^n$. Then $d(x,A) \le d_{\mathrm{C}}(x,A)$, where d is the Euclidean distance in $\mathbb{R}^n$.*

Proof. Let $w \in V_A(x)$ be such that $\|w\|_2 = \min_{v\in V_A(x)} \|v\|_2$. We can write $w = \sum_{u\in U_A(x)} \lambda_u u$, where every $\lambda_u \ge 0$ and $\sum_{u\in U_A(x)} \lambda_u = 1$. By definition, for each $u \in U_A(x)$, there exists $z^u \in A - x$ such that $z_i = 0$ for all i with $u_i = 0$. Since both A and x are contained in $[0,1]^n$, we have $|z_i^u| \le u_i$ for all i. Then $z = \sum_{u\in U_A(x)} \lambda_u z^u$ satisfies $|z_i| \le w_i$ for all i, and $d(x,A) = \inf_{y\in A-x} \|y\|_2 \le \|z\|_2 \le \|w\|_2 = d_{\mathrm{C}}(x,A)$. □

We can give an easy proof of a concentration result around the median.

Theorem 3.18 (Talagrand's inequality)**.** *Let $X = (X_1, \cdots, X_n)$ be a n-dimensional random vector with independent components taking values in $[0,1]$. If $F : [0,1]^n \to \mathbb{R}$ is a convex 1-Lipschitz function, and $M_{F(X)} := \inf\{x \in \mathbb{R} : \mathbb{P}(F(X) \le x) \ge 1/2\}$ the median of $F(X)$. Then for every $\epsilon > 0$,*

$$\mathbb{P}(|F(X) - M_{F(X)}| \ge \epsilon) \le 4e^{-\epsilon^2/4}.$$

Proof. By Theorem 3.15 and Lemma 3.17,

$$\mathbb{E}\left[\exp\left(\frac{d(X,A)^2}{4}\right)\right] \le \frac{1}{\mathbb{P}(X \in A)}.$$

Let $A = \{x \in \mathbb{R}^n : F(x) \le \alpha\}$, which is a closed convex set. By the Lipschitz property, if $F(x) \ge \alpha + \epsilon$ for some $\epsilon > 0$, then $d(x,A) \ge \epsilon$. Hence

$$\mathbb{P}(F(X) \ge \alpha + \epsilon) \le \mathbb{P}\left(d(X,A) \ge \epsilon\right) \le e^{-\epsilon^2/4}\mathbb{E}\left[\exp\left(\frac{d(X,A)^2}{4}\right)\right] \le \frac{e^{-\epsilon^2/4}}{\mathbb{P}(F(X) \le \alpha)}. \tag{3.9}$$

If $M_{F(X)}$ is the median of $F(X)$, then $\mathbb{P}(F(x) \le M_{F(X)}) \ge 1/2$ and $\mathbb{P}(F(X) < M_{F(X)}) \le 1/2$. Letting $\alpha = M_{F(X)}$ in (3.9), we get

$$\mathbb{P}\left(F(X) \ge M_{F(X)} + \epsilon\right) \le \frac{e^{-\epsilon^2/4}}{\mathbb{P}\left(F(x) \le M_{F(X)}\right)} \le 2e^{-\epsilon^2/4}.$$

Letting $\alpha = M_{F(X)} - \epsilon$ in (3.9), we get

$$\mathbb{P}\left(F(X) \leq M_{F(X)} - \epsilon\right) \leq \frac{e^{-\epsilon^2/4}}{\mathbb{P}\left(F(x) \geq M_{F(X)}\right)} \leq 2e^{-\epsilon^2/4}.$$

Thus we conclude the proof. $\square$

Remark. If $X = (X_1, \cdots, X_n)$ has independent components such that $\mathbb{P}(X_i \in [a_i, b_i] = 1)$ for each i, and $F : \mathbb{R}^n \to \mathbb{R}$ is a convex L-Lipschitz function, then

$$\mathbb{P}\left(|F(X) - M_{F(X)}| \geq \epsilon\right) \leq 4 \exp\left(-\frac{\epsilon}{4L^2 \max_{i\in[n]} |b_i - a_i|^2}\right).$$

One can get this result by rescaling F.

4 Tail Bound for Quadratic Forms

4.1 Gaussian Quadratic Forms

We first introduce the following lemma, which is used for controling the MGF of Gaussian quadratic forms.

Lemma 4.1 (Hsu et al., 2012, [3])**.** *Let $Z_1, \cdots, Z_m$ be independent standard Gaussian variables. Fix non-negative vector $\alpha \in \mathbb{R}_+^m$ and vector $\beta \in \mathbb{R}^m$. If $0 \le t < \frac{1}{2\|\alpha\|_\infty}$, then*

$$\mathbb{E}\left[\exp\left(t\sum_{i=1}^m \alpha_i Z_i^2 + \sum_{i=1}^m \beta_i Z_i\right)\right] \le \exp\left(t\|\alpha\|_1 + \frac{t^2\|\alpha\|_2^2 + \|\beta\|_2^2/2}{1-2t\|\alpha\|_\infty}\right). \tag{4.1}$$

Proof. Fix $0 \le t < 1/(2\|\alpha\|_\infty)$, and let $\eta_i = 1/\sqrt{1-2t\alpha_i} > 0$ for $i = 1, \cdots, m$. Then

$$\begin{aligned}
\mathbb{E}\left[\exp\left(t\alpha_i Z_i^2 + \beta_i Z_i\right)\right] &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \exp\left\{-\left(\frac{1}{2} - t\alpha_i\right)z^2 + \beta_i z\right\}\mathrm{d}z \\
&= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2}\left(\frac{z}{\eta_i} - \beta_i\eta_i\right)^2 + \frac{1}{2}\beta_i^2\eta_i^2\right\}\mathrm{d}z \\
&= \eta_i \exp\left(\frac{1}{2}\beta_i^2\eta_i^2\right) \\
&= \exp\left[-\frac{1}{2}\log(1-2t\alpha_i) + \frac{\beta_i^2}{2(1-2t\alpha_i)}\right].
\end{aligned} \tag{4.2}$$

Note that

$$-\log(1-2t\alpha_i) = \sum_{k=1}^{\infty}\frac{(2t\alpha_i)^k}{k} \le 2t\alpha_i + \sum_{k=2}^{\infty}\frac{(2t\alpha_i)^k}{2} = 2t\alpha + \frac{2t^2\alpha_i^2}{1-2t\alpha_i}. \tag{4.3}$$

Combining (4.2) and (4.3), we have

$$\begin{aligned}
\mathbb{E}\left[\exp\left(t\alpha_i Z_i^2 + \beta_i Z_i\right)\right] &\le \exp\left(t\alpha_i + \frac{t^2\alpha_i^2 + \beta_i^2/2}{1-2t\alpha_i}\right) \\
&\le \exp\left(t\alpha_i + \frac{t^2\alpha_i^2 + \beta_i^2/2}{1-2t\|\alpha\|_\infty}\right), \quad i = 1, \cdots, m.
\end{aligned} \tag{4.4}$$

Since $Z_1, \cdots, Z_m$ are independent, we have

$$\mathbb{E}\left[\exp\left(t\sum_{i=1}^m \alpha_i Z_i^2 + \sum_{i=1}^m \beta_i Z_i\right)\right] = \prod_{i=1}^m \mathbb{E}\left[\exp\left(t\alpha_i Z_i^2 + \beta_i Z_i\right)\right] \le \exp\left(t\|\alpha\|_1 + \frac{t^2\|\alpha\|_2^2 + \|\beta\|_2^2/2}{1-2t\|\alpha\|_\infty}\right).$$

Then we complete the proof. □

Remark. Using a similar argument, for every $t > 0$, we also have

$$\mathbb{E}\left[\exp\left(-t\alpha_i Z_i^2 + \beta_i Z_i\right)\right] \le \exp\left[-\frac{1}{2}\log(1+2t\alpha_i) + \frac{\beta_i^2}{2(1+2t\alpha_i)}\right].$$

Using the inequality $\log(1+x) \ge x - \frac{x^2}{2}$, we have

$$\mathbb{E}\left[\exp\left(-t\sum_{i=1}^m \alpha_i Z_i^2\right) + \sum_{i=1}^m \beta_i Z_i\right] \le \exp\left(-t\|\alpha\|_1 + t^2\|\alpha\|_2^2 + \frac{1}{2}\|\beta\|_2^2\right), \quad t \ge 0. \tag{4.5}$$

This is a lower bound for the MGF of Gaussian quadratic form.

4.2 Positive-Definite Quadratic Forms of Sub-Gaussian Variables

In this subsection we introduce a tail bound for the quadratic form of sub-Gaussian vectors.

Theorem 4.2 (Laurent-Massart-Hsu, [3])**.** *Let $X_1, \cdots, X_n$ be independent sub-Gaussian variables with mean 0 and variance proxy σ^2, and fix any positive definite matrix $\Sigma \in \mathbb{R}^{n\times n}$. Then for each $t > 0$, we have*

$$\mathbb{P}\left(X^\top \Sigma X \geq \sigma^2 \left(\mathrm{tr}(\Sigma) + 2\|\Sigma\|_{\mathrm{F}}\sqrt{t} + 2\|\Sigma\|_2\, t\right)\right) \leq e^{-t}, \tag{4.6}$$

where $X = (X_1, \cdots, X_n)^\top$. Furthermore, if $X_1, \cdots, X_n$ are Gaussian, then for each $t > 0$,

$$\mathbb{P}\left(X^\top \Sigma X \leq \sigma^2 \left(\mathrm{tr}(\Sigma) - 2\|\Sigma\|_{\mathrm{F}}\sqrt{t}\right)\right) \leq e^{-t}, \tag{4.7}$$

Proof. Since Σ is positive definite, it admits a spectral decomposition $\Sigma = Q^\top S Q$ where $Q \in \mathbb{R}^{n\times n}$ is an orthogonal matrix and $S = \mathrm{diag}\{\rho_1, \cdots, \rho_n\}$ with eigenvalues $\rho_1 \geq \cdots \geq \rho_n > 0$. Let Z be a vector of n independent standard Gaussian variables. Then for any $\alpha \in \mathbb{R}^n$ and $\epsilon > 0$, we have

$$\mathbb{E}\left[e^{Z^\top \alpha}\right] = \prod_{j=1}^{n} \mathbb{E}[e^{Z_j \alpha_j}] = e^{\|\alpha\|_2^2/2}. \tag{4.8}$$

Denote $A = Q^\top S^{1/2} Q$. For any $\epsilon > 0$, define set $E_\epsilon = \left\{x \in \mathbb{R}^n : x^\top \Sigma x \geq \epsilon\right\}$. Fix $\lambda > 0$, we have

$$\begin{aligned}\mathbb{E}\left[\exp\left(\lambda Z^\top A X\right)\right] &= \int_{\mathbb{R}^n} \mathbb{E}\left[\exp\left(\lambda Z^\top A X\right) | X = x\right] \mathrm{d}F_X(x) = \int_{\mathbb{R}^n} \exp\left(\frac{1}{2}\lambda^2 x^\top \Sigma x\right) \mathrm{d}F_X(x) \\ &\geq \int_{E_\epsilon} \exp\left(\frac{1}{2}\lambda^2 x^\top \Sigma x\right) \mathrm{d}F_X(x) \geq \exp\left(\frac{1}{2}\lambda^2 \epsilon\right) \mathbb{P}(X^\top \Sigma X \geq \epsilon),\end{aligned} \tag{4.9}$$

where the second equality follows from (4.8). Moreover, by sub-Gaussianity of X,

$$\mathbb{E}\left[\exp\left(\lambda Z^\top A X\right)\right] = \mathbb{E}\left[\mathbb{E}\left[\exp\left(\lambda Z^\top A X\right) | Z\right]\right] \leq \mathbb{E}\left[\exp\left(\frac{\lambda^2\sigma^2}{2} Z^\top \Sigma Z\right)\right]. \tag{4.10}$$

Combining (4.9) and (4.10) gives

$$\mathbb{P}(X^\top \Sigma X \geq \epsilon) \leq \mathbb{E}\left[\exp\left(\frac{\lambda^2\sigma^2}{2} Z^\top \Sigma Z - \frac{1}{2}\lambda^2 \epsilon\right)\right].$$

Define $Y = QZ$, the orthogonality of Q implies that Y is also a vector of n independent standard Gaussian variables, and $Z^\top \Sigma Z = Y^\top S Y = \sum_{i=1}^n \rho_i Y_i^2$. Let $\rho = (\rho_1, \cdots, \rho_n)^\top$ and $\gamma = \lambda^2\sigma^2/2$. By Lemma 4.1, we may choose λ to ensure that $0 \leq \gamma < 1/(2\|\rho\|_\infty)$, and so

$$\mathbb{P}(X^\top \Sigma X \geq \epsilon) \leq \exp\left(-\frac{\gamma\epsilon}{\sigma^2} + \gamma\|\rho\|_1 + \frac{\gamma^2\|\rho\|_2^2}{1 - 2\gamma\|\rho\|_\infty}\right).$$

Let $\delta = 1 - 2\gamma\|\rho\|_\infty$ with $0 < \delta \leq 1$. Then

$$\mathbb{P}(X^\top \Sigma X \geq \epsilon) \leq \exp\left\{\frac{1}{2\|\rho\|_\infty}\left[(1-\delta)\left(\|\rho\|_1 - \frac{\epsilon}{\sigma^2}\right) + \frac{\|\rho\|_2^2}{2\|\rho\|_\infty}\left(\delta + \delta^{-1} - 2\right)\right]\right\}.$$

Let $\epsilon = \sigma^2 \left[\|\rho\|_1 + \frac{\|\rho\|_2^2}{2\|\rho\|_\infty}(\delta^{-2} - 1)\right]$. Then we have

$$\mathbb{P}\left(X^\top \Sigma X \geq \sigma^2 \left[\|\rho\|_1 + \frac{\|\rho\|_2^2}{2\|\rho\|_\infty}\left(\frac{1}{\delta^2} - 1\right)\right]\right) \leq \exp\left[-\frac{\|\rho\|_2^2}{4\|\rho\|_\infty^2}\left(\frac{1}{\delta} - 1\right)^2\right].$$

Now let $t = \frac{\|\rho\|_2^2}{4\|\rho\|_\infty^2}\left(\delta^{-1} - 1\right)^2 \geq 0$, that is, $\delta^{-1} = 1 + \frac{2\|\rho\|_\infty}{\|\rho\|_2}\sqrt{t}$, then

$$\mathbb{P}\left(X^\top \Sigma X \geq \sigma^2 \left(\|\rho\|_1 + 2\|\rho\|_2\sqrt{t} + 2\|\rho\|_\infty t\right)\right) \leq e^{-t}. \tag{4.11}$$

Recall that $\rho_1, \cdots, \rho_n$ are eigenvalues of Σ, we have $\|\rho\|_1 = \mathrm{tr}(\Sigma)$, $\|\rho\|_2 = \|\Sigma\|_{\mathrm{F}}$ and $\|\rho\|_\infty = \|\Sigma\|_2$, and the result (4.6) immediately follows from (4.11). For the proof of (4.7), we assume that $X_1, \cdots, X_n$ are Gaussian. Then similar to (4.9), for any $\epsilon > 0$,

$$\mathbb{E}\left[\exp\left(i\lambda Z^\top AX\right)\right] \geq \exp\left(-\frac{1}{2}\lambda^2\epsilon\right)\mathbb{P}(X^\top \Sigma X \leq \epsilon).$$

Meanwhile, due to Gaussianity of $X_1, \cdots, X_n$, (4.10) becomes

$$\mathbb{E}\left[\exp\left(i\lambda Z^\top AX\right)\right] = \mathbb{E}\left[\exp\left(-\frac{\lambda^2\sigma^2}{2}Z^\top \Sigma Z\right)\right]$$

Hence

$$\mathbb{P}(X^\top \Sigma X \leq \epsilon) \leq \mathbb{E}\left[\exp\left(\frac{1}{2}\lambda^2\epsilon - \frac{\lambda^2\sigma^2}{2}Z^\top \Sigma Z\right)\right].$$

Using the inequality (4.5), with $\gamma = \lambda^2\sigma^2/2$, we have

$$\mathbb{P}(X^\top \Sigma X \leq \epsilon) \leq \exp\left(\frac{\gamma\epsilon}{\sigma^2} - \gamma\|\rho\|_1 + \gamma^2\|\rho\|_2^2\right).$$

Let $\epsilon = \sigma^2(\|\rho\|_1 - 2\gamma\|\rho\|_2^2)$ and $t = \gamma^2\|\rho\|_2^2$. Then

$$\mathbb{P}\left(X^\top \Sigma X \leq \sigma^2 \left(\|\rho\|_1 - 2\|\rho\|_2\sqrt{t}\right)\right) \leq e^{-t}.$$

Then we conclude the proof. □

The following corollary immediately holds by setting Σ in Theorem 4.2 as the $n \times n$ identity matrix.

Corollary 4.3. *Let $X_1, \cdots, X_n$ be independent sub-Gaussian variables with mean 0 and variance proxy σ^2. Then for every $t > 0$, we have*

$$\mathbb{P}\left(\sum_{i=1}^n X_i^2 \geq \sigma^2\left(n + 2\sqrt{nt} + 2t\right)\right) \leq e^{-t}.$$

Furthermore, if $X_1, \cdots, X_n$ are Gaussian, we have

$$\mathbb{P}\left(\sum_{i=1}^n X_i^2 \leq \sigma^2\left(n - 2\sqrt{nt}\right)\right) \leq e^{-t}.$$

This is a sharper tail bound for the chi-square distribution.

4.3 Application I: Ordinary Least Square with A Fixed Design

We consider a fixed dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N \subset \mathbb{R}^d \times \mathbb{R}$ from a linear model: $y_i = x_i^\top \beta^* + \epsilon_i$, where $\beta^* \in \mathbb{R}^d$ and $\{\epsilon_i\}_{i=1}^N$ are independent σ^2-sub-Gaussian noises with $\mathbb{E}\epsilon_i = 0$. The solution to the ordinary least square (OLS) problem is

$$\widehat{\beta} = \operatorname*{argmin}_{\beta \in \mathbb{R}^d} \sum_{i=1}^N (y_i - x_i^\top \beta)^2 = \Sigma^{-1} \left(\sum_{i=1}^N x_i y_i \right),$$

where $\Sigma = \sum_{i=1}^N x_i x_i^\top \in \mathbb{R}^{d \times d}$. In many cases, we are interested in the difference between our estimator $\widehat{\beta}$ and the true parameter β^*.

Proposition 4.4 (OLS with a fixed design)**.** *Assume that Σ is invertible, and $0 < \delta < 1$. With probability at least $1 - \delta$, we have*

$$\|\widehat{\beta} - \beta^*\|_\Sigma^2 \le \sigma^2 \left(d + 2\sqrt{d \log(1/\delta)} + 2\log(1/\delta) \right).$$

Proof. Denote by $X = (x_1, \cdots, x_N)^\top \in \mathbb{R}^{N \times d}$ the covariate matrix, $\epsilon = (\epsilon_1, \cdots, \epsilon_N)^\top$ the noise vector, and $Y = (y_1, \cdots, y_N)^\top = X\beta^* + \epsilon$ the response vector. Then we have $\Sigma = X^\top X$, and

$$\|\widehat{\beta} - \beta^*\|_\Sigma^2 = (\Sigma^{-1} X^\top Y - \beta^*)^\top \Sigma (\Sigma^{-1} X^\top Y - \beta^*) = \epsilon^\top X \Sigma^{-1} X^\top \epsilon.$$

Note that ϵ is σ^2-sub-Gaussian, $\operatorname{tr}(X\Sigma^{-1}X^\top) = d$, $\|X\Sigma^{-1}X^\top\|_{\mathrm{F}}^2 = d$ and $\|X\Sigma^{-1}X^\top\|_2 = 1$, we can apply Theorem 4.2 to any $t > 0$:

$$\mathbb{P}\left(\epsilon^\top X \Sigma^{-1} X^\top \epsilon \ge \sigma^2 \left(d + 2\sqrt{dt} + 2t \right) \right) \le e^{-t}. \tag{4.12}$$

Then the result immediately follows from (4.12) by setting $t = \log(1/\delta)$. □

4.4 Application II: Johnson-Lindestrauss Lemma

Theorem 4.5 (Johnson-Lindestrauss)**.** *Let $0 < \epsilon < 1/2$, $G = (G_{ij})_{d \times n}$ a matrix with i.i.d. standard Gaussian entries, and $A = G/\sqrt{d}$. If the dimension*

$$d \ge \frac{2}{\epsilon^2} \log \frac{2N}{\delta},$$

then

$$\mathbb{P}\left(1 - \epsilon < \frac{\|Ax\|_2}{\|x\|_2} < 1 + \epsilon \text{ for all } x \in \mathcal{X}_N \right) \ge 1 - \delta.$$

Proof. We first note that for every $x \in \mathbb{R}^n \backslash \{0\}$, Ax is jointly Gaussian, and

$$\mathbb{E}\|Au - Av\|_2^2 = \mathbb{E}\left| \sum_{j=1}^n A_{ij} x_j \right|^2 = \frac{\|x\|_2^2}{d}.$$

Hence $Au - Av$ has sub-Gaussian elements with proxy $\|u - v\|_2^2/d$. Then by Corollary 4.3,

$$\mathbb{P}\left(\frac{\|Ax\|_2^2}{\|x\|_2^2} \ge 1 + 2\sqrt{\frac{t}{d}} + \frac{2t}{d} \right) \le e^{-t}, \quad t > 0,$$

and

$$\mathbb{P}\left(\frac{\|Ax\|_2}{\|x\|_2} \le 1 - 2\sqrt{\frac{t}{d}} \right) \le e^{-t}, \quad t > 0.$$

We take $t = d\epsilon^2/2$. The above two displays together imply

$$\mathbb{P}\left(1 - \sqrt{2}\epsilon < \frac{\|Ax\|_2^2}{\|x\|_2^2} < 1 + \sqrt{2}\epsilon + \epsilon^2 \right) \ge 1 - 2e^{-d\epsilon^2/2}$$

Since $0 < \epsilon < 1/2 < 2 - \sqrt{2}$, we have $(1-\epsilon)^2 < 1 - \sqrt{2}\epsilon < 1 + \sqrt{2}\epsilon + \epsilon^2 < (1+\epsilon)^2$. An application of the union bound gives

$$\mathbb{P}\left(1 - \epsilon < \frac{\|Ax\|_2}{\|x\|_2} < 1 + \epsilon \text{ for all } x \in \mathcal{X}_N\right) \geq 1 - 2Ne^{-d\epsilon^2/2}$$

Setting $d \geq 2\epsilon^{-2}\log(2N/\delta)$ in the above inequality proves the result. □

We can extend this result to the whole space.

Theorem 4.6. *Let $0 < \epsilon < 1/2$, $G = (G_{ij})_{d\times n}$ a matrix with i.i.d. standard Gaussian entries, and $A = G/\sqrt{d}$. If the dimension*

$$d \geq \frac{18}{\epsilon^2}\left[n\log\left(1 + \frac{6}{\epsilon}\right) + \log\frac{2}{\delta}\right],$$

then

$$\mathbb{P}\left(1 - \epsilon < \frac{\|Ax\|_2}{\|x\|_2} < 1 + \epsilon \textit{ for all } x \in \mathbb{R}^n\right) \geq 1 - \delta.$$

Proof. Step I. Fix $0 < \epsilon < 1/2$. We first construct an ϵ-net $\mathcal{N}_\epsilon$ of the unit sphere $S^{n-1} := \{x \in \mathbb{R}^n : \|x\|_2 = 1\}$ by a greedy algorithm:

- Initialize $\mathcal{N}_\epsilon = \{x_1\}$, where $x_1 \in S^{n-1}$ is arbitrary;
- While such a point exists, choose an arbitrary point $y \in S^{n-1}$ such that $\|y - x\|_2 > \epsilon$ for every $x \in \mathcal{N}_\epsilon$, and update $\mathcal{N}_\epsilon \leftarrow \mathcal{N}_\epsilon \cup \{y\}$.

If this algorithm stops after finite steps, it automatically produces an ϵ-net $\mathcal{N}_\epsilon$. Indeed, after N points are added to $\mathcal{N}_\epsilon$, one have $\|x_j - x_k\|_2 > \epsilon$ for every pair of distinct $x_j, x_k \in \mathcal{N}_\epsilon$, otherwise either x_j or x_k would not have been added to $\mathcal{N}_\epsilon$. Consequently, the balls $\{B(x_k, \epsilon/2)\}_{k=1}^N$ are pairwise disjoint and contained in the ball $B(0, 1 + \epsilon/2)$, and

$$\frac{N\epsilon^n}{2^n} \leq \left(1 + \frac{\epsilon}{2}\right)^n.$$

Hence the greedy algorithm will stop after finite steps, and $|\mathcal{N}_\epsilon| \leq \left(1 + \frac{2}{\epsilon}\right)^n$.

Step II. For every $x \in S^{n-1}$, we can find $x_0 \in \mathcal{N}_\epsilon$ such that $\|x - x_0\| \leq \epsilon$. Next, we can also find $x_1 \in \mathcal{N}_\epsilon$ that approximates normalized $x - x_0$ with error no greater that ϵ. Repeating this procedure, every $x \in S^{n-1}$ can be represented as

$$x = x_0 + \sum_{k=1}^{\infty} c_k x_k,$$

where $x_0, x_1, \cdots \in \mathcal{N}_\epsilon$, the coefficients $c_k \leq \epsilon^k$ for all $k \in \mathbb{N}$, and the convergence holds in $\|\cdot\|_2$ norm. Then for every $A \in \mathbb{R}^{d\times n}$,

$$\|Ax\|_2 \leq \sum_{k=0}^{n} \epsilon^k \|Ax_k\|_2 \leq \sup_{y\in\mathcal{N}_\epsilon} \frac{\|Ay\|_2}{1-\epsilon},$$

and

$$\|Ax\|_2 \geq \|Ax_0\|_2 - \sum_{k=1}^{n} c_k\|Ax_k\|_2 \geq \|Ax_0\|_2 - \sum_{k=1}^{n} \epsilon^k\|Ax_k\|_2 \geq \inf_{y\in\mathcal{N}_\epsilon} \|Ay\|_2 - \sup_{y\in\mathcal{N}_\epsilon} \frac{\epsilon\,\|Ay\|_2}{1-\epsilon}.$$

Step III. By the Johnson-Linderstrauss lemma,

$$\mathbb{P}\left(1 - \epsilon < \|Ax\|_2 < 1 + \epsilon \text{ for all } x \in \mathcal{N}_\epsilon\right) \geq 1 - 2\,|\mathcal{N}_\epsilon|\, e^{-d\epsilon^2/2}.$$

On the above event,

$$1 - 3\epsilon \leq \frac{1-3\epsilon}{1-\epsilon} \leq \|Ax\|_2 \leq \frac{1+\epsilon}{1-\epsilon} \leq 1 + 3\epsilon.$$

Therefore

$$\mathbb{P}\left(1 - 3\epsilon < \|Ax\|_2 < 1 + 3\epsilon \text{ for all } x \in S^{n-1}\right) \geq 1 - 2\left(1 + \frac{2}{\epsilon}\right)^n e^{-d\epsilon^2/2}.$$

In order that the right-hand side of the above bound is less than $1-\delta$, we let

$$d \geq \frac{2}{\epsilon^2}\left[n\log\left(1+\frac{2}{\epsilon}\right)+\log(2/\delta)\right],$$

Finally, replacing ϵ by $\epsilon/3$ completes the proof. $\square$

4.5 Hanson-Wright Inequality

Theorem 4.7 (Hanson-Wright inequality)**.** *Let $A \in \mathbb{R}^{n\times n}$. Let $X = (X_1, \cdots, X_n)$ be a random vector with independent, mean-zero, subgaussian components with proxy σ^2. Then for every $\epsilon > 0$,*

$$\mathbb{P}\left(\left|X^\top AY - \mathbb{E}\left[X^\top AY\right]\right| \geq \epsilon\right) \leq 4\exp\left(-c\min\left\{\frac{\epsilon^2}{\sigma^4\|A\|_{\mathrm{F}}^2}, \frac{\epsilon}{\sigma^2\|A\|_2}\right\}\right),$$

where $c > 0$ is an absolute constant.

Proof. We use the following decomposition:

$$X^\top AX - \mathbb{E}\left[X^\top AX\right] = \sum_{i=1}^n a_{ii}(X_i^2 - \mathbb{E}[X_i^2]) + \sum_{i\neq j}^n a_{ij}X_iX_j.$$

Step I. Since X_i are independent and subgaussian, the variables $X_i^2 - \mathbb{E}[X_i^2]$ are independent, mean-zero, and subexponential. By Example 1.15, for all $|t| \leq 1/(4\sigma^2\max_i|a_{ii}|)$, the diagonal sum satisfies

$$\begin{aligned}\mathbb{E}\left[\exp\left(t\sum_{i=1}^n a_{ii}(X_i^2-\mathbb{E}[X_i^2])\right)\right] &= \prod_{i=1}^n \mathbb{E}\left[\exp\left(ta_{ii}(X_i^2-\mathbb{E}[X_i^2])\right)\right] \\ &\leq \prod_{i=1}^n\left(1+32t^2a_{ii}^2\sigma^4\right) = \exp\left(32\sum_{i=1}^n a_{ii}^2t^2\sigma^4\right) \leq \exp\left(32t^2\sigma^4\|A\|_{\mathrm{F}}^2\right).\end{aligned}$$

Note that $\max_i a_{ii}^2 \leq \|A\|_2^2$. By Theorem 1.11 (ii),

$$\mathbb{P}\left(\left|\sum_{i=1}^n a_{ii}(X_i^2-\mathbb{E}[X_i^2])\right| \geq \epsilon\right) \leq 2\exp\left(-\frac{1}{2}\min\left\{\frac{\epsilon^2}{64\sigma^4\|A\|_{\mathrm{F}}^2}, \frac{\epsilon}{4\sigma^2\|A\|_2}\right\}\right).$$

To handle the off-diagonal sum, we need some technical lemmas.

Lemma 4.8 (Decoupling)**.** *Let $A \in \mathbb{R}^{n\times n}$ be a diagonal-free matrix (i.e. all diagonal entries of A equal zero). Let X be a zero-mean random vector in $\mathbb{R}^n$, and let Y be an independent copy of X. Then*

$$\mathbb{E}\left[F(X^\top AX)\right] \leq \mathbb{E}\left[F(4X^\top AY)\right].$$

Proof. Step I. We let $\epsilon_1, \cdots, \epsilon_n \sim \mathrm{Bernoulli}(1/2)$ be i.i.d. Bernoulli variables that are independent of X and Y. Let $I = \{i \in [n] : \epsilon_i = 1\}$, which is a random set. Then $\mathbb{E}\left[\epsilon_i(1-\epsilon_j)\right] = \frac{1}{4}$ for all $i \neq j$, and

$$\sum_{i\neq j}^n a_{ij}X_iX_j = 4\mathbb{E}_\epsilon\sum_{i\neq j}\epsilon_i(1-\epsilon_j)a_{ij}X_iX_j = 4\mathbb{E}_I\left[\sum_{i\in I}\sum_{j\in I^c}a_{ij}X_iX_j\right]$$

Then by Jensen's inequality and Fubini's theorem,

$$\mathbb{E}\left[F(X^\top AX)\right] = \mathbb{E}_X F\left(4\mathbb{E}_I\left[\sum_{i\in I}\sum_{j\in I^c} a_{ij}X_iX_j\right]\right) \le \mathbb{E}_I\mathbb{E}_X F\left(4\sum_{i\in I}\sum_{j\in I^c} a_{ij}X_iX_j\right)$$

Since I and I^c are disjoint, we replace X_j by Y_j for $j \in I$:

$$\mathbb{E}\left[F(X^\top AX)\right] \le \mathbb{E}F\left(4\sum_{i\in I}\sum_{j\in I^c} a_{ij}X_iY_j\right). \tag{4.13}$$

Step II. Note that

$$X^\top AY = \sum_{i,j\in[n]} a_{ij}X_iY_j = \sum_{i\in I}\sum_{j\in I^c} a_{ij}X_iY_j + \sum_{i\in I}\sum_{j\in I} a_{ij}X_iY_j + \sum_{i\in I^c}\sum_{j\in[n]} a_{ij}X_iY_j$$

If we condition on $(X_i)_{i\in I}$ and $(Y_j)_{j\in I^c}$, we have

$$\mathbb{E}\left[\sum_{i\in I}\sum_{j\in I} a_{ij}X_iY_j + \sum_{i\in I^c}\sum_{j\in[n]} a_{ij}X_iY_j \,\middle|\, (X_i)_{i\in I}, (Y_j)_{j\in I^c}\right] = 0.$$

By conditional Jensen's inequality,

$$\mathbb{E}\left[4F(X^\top AY) \,|\, (X_i)_{i\in I}, (Y_j)_{j\in I^c}\right] \ge F\left(4\sum_{i\in I}\sum_{j\in I^c} a_{ij}X_iY_j\right).$$

Taking expectation on both sides and comparing with (4.13), we obtain the desired result. □

Lemma 4.9 (Gaussian replacement)**.** *Let $A \in \mathbb{R}^{n\times n}$, and let $X_1, \cdots, X_n$ be sub-Gaussian random variables with proxy $\sigma_1^2, \cdots, \sigma_n^2$, and let $Y_1, \cdots, Y_n$ are independent copies of $X_1, \cdots, X_n$. Also, let $\xi, \eta \in N(0, I_n)$ be independent Gaussian vectors. Then*

$$\mathbb{E}\left[\exp\left(tX^\top AY\right)\right] \le \mathbb{E}\left[t\sigma^2\xi^\top A\eta\right], \quad t \in \mathbb{R}.$$

Proof. By sub-Gaussianity, we have

$$\mathbb{E}_X\left[\exp\left(tX^\top AY\right)\right] \le \exp\left(\frac{t^2\sigma^2\|AY\|_2^2}{2}\right), \quad t \in \mathbb{R}.$$

For the Gaussian random vector $\xi \sim N(0, I_n)$,

$$\mathbb{E}_\xi\left[\exp\left(s\,\xi^\top AY\right)\right] = \exp\left(\frac{s^2\|AY\|_2^2}{2}\right), \quad s \in \mathbb{R}.$$

Setting $s = t\sigma$, we match the right-hand sides of the above two estimates to obtain

$$\mathbb{E}_X\left[\exp\left(tX^\top AY\right)\right] \le \mathbb{E}_\xi\left[\exp\left(t\sigma\xi^\top AY\right)\right]. \tag{4.14}$$

Repeating this procedure, for $\eta \in N(0, I_n)$, we have

$$\mathbb{E}_Y\left[\exp\left(t\sigma\xi^\top AY\right)\right] \le \mathbb{E}_\eta\left[\exp\left(t\sigma^2\xi A^\top\eta\right)\right]. \tag{4.15}$$

Combining (4.14) and (4.15), and by Fubini's theorem,

$$\mathbb{E}_Y\left[\mathbb{E}_X\left[\exp\left(tX^\top AY\right)\right]\right] \le \mathbb{E}_\xi\left[\mathbb{E}_\eta\left[\exp\left(t\sigma^2\xi A^\top\eta\right)\right]\right].$$

Then we get the desired result. □

Lemma 4.10. *Let $A \in \mathbb{R}^{n\times n}$, and let $\xi, \eta \in N(0, I_n)$ be independent. Then*

$$\mathbb{E}\left[\exp\left(t\xi^\top A\eta\right)\right] \le \exp\left(t^2\|A\|_{\mathrm{F}}^2\right), \quad \textit{for all } |t| \le \frac{1}{2\|A\|_2}$$

Proof. We consider the SVD $A = U\Sigma V^\top$, where $\Sigma = \mathrm{diag}(s_1, \cdots, s_n)$, where $s_1 \ge \cdots \ge s_n \ge 0$. Since $N(0, I_n)$ is rotation-invariant, we do diagonalization:

$$\xi^\top A\eta = \xi^\top U\Sigma V^\top\eta = (U^\top\xi)^\top\Sigma(V^\top\eta) \overset{d}{=} \xi^\top\Sigma\eta = \sum_{i=1}^n s_i\xi_i\eta_i.$$

This is a sum of independent random variables, and

$$\mathbb{E}\left[\exp\left(t\xi^\top A\eta\right)\right] = \prod_{i=1}^n \mathbb{E}\left[e^{ts_i\xi_i\eta_i}\right] = \prod_{i=1}^n \mathbb{E}\left[\exp\left(\frac{t^2s_i^2\xi_i^2}{2}\right)\right], \quad t \in \mathbb{R}.$$

If $t^2 \le 1/(2\max_i s_i^2)$, we have

$$\mathbb{E}\left[\exp\left(\frac{t^2s_i^2\xi_i^2}{2}\right)\right] \le \frac{1}{\sqrt{1-t^2s_i^2}} \le e^{t^2s_i^2}.$$

Then we have

$$\mathbb{E}\left[\exp\left(t\xi^\top A\eta\right)\right] \le \exp\left(t^2\sum_{i=1}^n s_i^2\right) = \exp\left(t^2\|A\|_{\mathrm{F}}^2\right),$$

and we finish the proof. □

Step II. By decoupling [Lemma 4.8], Gaussian replacement [Lemma 4.9] and Lemma 4.10, for each $t \in \mathbb{R}$ with $|t| \le 1/(8\sigma^2\|A\|_2)$, the off-diagonal sum satisfies

$$\begin{aligned}\mathbb{E}\left[\exp\left(t\sum_{i\ne j}^n a_{ij}X_iX_j\right)\right] &\le \mathbb{E}\left[\exp\left(4t\sum_{i\ne j}^n a_{ij}X_iY_j\right)\right]\\ &\le \mathbb{E}\left[\exp\left(4t\sigma^2\sum_{i\ne j}^n a_{ij}\xi_i\eta_j\right)\right] \le \exp\left(16t^2\sigma^4\|A\|_{\mathrm{F}}^2\right).\end{aligned}$$

By Theorem 1.11 (ii),

$$\mathbb{P}\left(\left|\sum_{i\ne j}^n a_{ij}X_iX_j\right| \ge \epsilon\right) \le 2\exp\left(-\frac{1}{2}\min\left\{\frac{\epsilon^2}{32\sigma^4\|A\|_{\mathrm{F}}^2}, \frac{\epsilon}{8\sigma^2\|A\|_2}\right\}\right).$$

Step III. Combining Steps I and II, we have

$$\mathbb{P}\left(\left|X^\top AX - \mathbb{E}[X^\top AX]\right| \ge 2\epsilon\right) \le 4\exp\left(-\frac{1}{2}\min\left\{\frac{\epsilon^2}{64\sigma^4\|A\|_{\mathrm{F}}^2}, \frac{\epsilon}{8\sigma^2\|A\|_2}\right\}\right).$$

Then we finish the proof by substituting ϵ with $\epsilon/2$ and setting $c = 1/512$. □

5 Application in Empirical process

5.1 Dudley's Entropy Integral

Definition 5.1 (Sub-Gaussian process)**.** Let $\{X_f : f \in \mathcal{F}\}$ be a collection of mean zero random variables indexed by $f \in \mathcal{F}$, and let d be a metric on the index set $\mathcal{F}$. Then $\{X_f : f \in \mathcal{F}\}$ is said to be a *sub-Gaussian process with respect to* d if

$$\mathbb{E}\left[e^{t(X_f - X_g)}\right] \leq \exp\left\{\frac{t^2 d^2(f,g)}{2}\right\}, \ \forall f, g \in \mathcal{F}.$$

That is, $X_f - X_g$ is sub-Gaussian with variance proxy $d^2(f,g)$.

Definition 5.2 (ϵ-covering number/metric entropy)**.** Let $(\mathcal{F}, d)$ be a metric space. For $\epsilon > 0$, a subset $\mathcal{N}_\epsilon \subseteq \mathcal{F}$ is called a ϵ-net of $\mathcal{F}$, if $\mathcal{F} \subseteq \bigcup_{f \in \mathcal{N}_\epsilon} B(f, \epsilon)$, where $B(f, \epsilon)$ is the open d-ball of radius ϵ centered at f. The *ϵ-covering number of* $\mathcal{F}$ is the cardinality of the minimal ϵ-cover of $\mathcal{F}$, i.e.

$$N(\epsilon, \mathcal{F}, d) = \min\{|\mathcal{N}_\epsilon|, \ \mathcal{N}_\epsilon \text{ is an } \epsilon\text{-net of } \mathcal{F}\}.$$

The following theorem can be seen as an extension of Theorem 3.3.

Theorem 5.3 (Dudley)**.** *Let $(\mathcal{F}, d)$ be a metric space, and suppose that $D := \sup_{f,g \in \mathcal{F}} d(f,g) < \infty$. Let $\{X_f : f \in \mathcal{F}\}$ be a stochastic process such that*

(i) $\{X_f, f \in \mathcal{F}\}$ is sub-Gaussian with respect to d, and

(ii) $\{X_f, f \in \mathcal{F}\}$ is sample-continuous, i.e., for each sequence $\{f_n\} \subset \mathcal{F}$ such that $\lim_{n\to\infty} d(f_n, f) = 0$ for some $f \in \mathcal{F}$, we have $X_{f_n} \to X_f$ almost surely.

Then the expected supremum of $\{X_f, f \in \mathcal{F}\}$ can be bounded with Dudley's entropy integral:

$$\mathbb{E}\left[\sup_{f \in \mathcal{F}} X_f\right] \leq 12 \int_0^{D/2} \sqrt{\log N(\epsilon, \mathcal{F}, d)}\, \mathrm{d}\epsilon. \tag{5.1}$$

Proof. This proof uses Dudley's chaining rule. Choose an arbitrary $f_0 \in \mathcal{F}$ and set $\epsilon_0 = D$, then $\mathcal{N}_{\epsilon_0} = \{f_0\}$ is a ϵ_0-net of $\mathcal{F}$. Now we choose a sequence of minimal ϵ-nets $\{\mathcal{N}_{\epsilon_j}\}$ by setting $\epsilon_j := 2^{-j}\epsilon_0$ for $j = 1, 2, \cdots$. For brevity, write $\mathcal{N}_j = \mathcal{N}_{\epsilon_j}$. By definition of ϵ-net, $\forall f \in \mathcal{F}$, we can find $f_j \in \mathcal{N}_j$ such that $d(f, f_j) \leq \epsilon_j$ for all $j \in \mathbb{N}$. Fixing $m \in \mathbb{N}$, we have

$$X_f = (X_f - X_{f_m}) + \sum_{j=1}^{m} \left(X_{f_j} - X_{f_{j-1}}\right) + X_{f_0}. \tag{5.2}$$

Note that both f_j and f_{j-1} are close to f, we have $d(f_j, f_{j-1}) = d(f_j, f) + d(f, f_{j-1}) \leq 3\epsilon_j$. Define a new class $\mathcal{H}_j = \{(g_{j-1}, g_j) \in \mathcal{N}_{j-1} \times \mathcal{N}_j : d(g_j, g_{j-1}) \leq 3\epsilon_j\}$, $j \in \mathbb{N}$. We have $|\mathcal{H}_j| \leq |\mathcal{N}_{j-1}||\mathcal{N}_j| \leq |\mathcal{N}_j|^2$.

Revisiting (4.4), we have

$$\begin{aligned}
\mathbb{E}\left[\sup_{f \in \mathcal{F}} X_f\right] &\leq \mathbb{E}\left[\sup_{g,g' \in \mathcal{F},\, d(g,g') \leq \epsilon_m} (X_g - X_{g'}) + \sum_{j=1}^{m} \max_{(g_{j-1}, g_j) \in \mathcal{H}_j} \left(X_{g_j} - X_{g_{j-1}}\right) + X_{f_0}\right] \\
&= \mathbb{E}\left[\sup_{g,g' \in \mathcal{F},\, d(g,g') \leq \epsilon_m} (X_g - X_{g'})\right] + \sum_{j=1}^{m} \mathbb{E}\left[\max_{(g_{j-1}, g_j) \in \mathcal{H}_j} \left(X_{g_j} - X_{g_{j-1}}\right)\right],
\end{aligned} \tag{5.3}$$

where the equality follows from $\mathbb{E}[X_{f_0}] = 0$. Since $\{X_{g_j} - X_{g_{j-1}}, \ (g_{j-1}, g_j) \in \mathcal{H}_j\}$ are sub-Gaussian with

variance proxy $9\epsilon_j^2$, it follows from applying Theorem 3.3 that

$$\mathbb{E}\left[\max_{(g_{j-1},g_j)\in\mathcal{H}_j}(X_{g_j}-X_{g_{j-1}})\right] \le 3\epsilon_j\sqrt{2\log|\mathcal{H}_j|} \le 6\epsilon_j\sqrt{\log|\mathcal{N}_j|} = 12(\epsilon_j-\epsilon_{j+1})\sqrt{\log N(\epsilon_j,\mathcal{F},d)}. \tag{5.4}$$

Plug in (5.4) to (5.3), we have

$$\begin{aligned}
\mathbb{E}\left[\sup_{f\in\mathcal{F}} X_f\right] &\le \mathbb{E}\left[\sup_{g,g'\in\mathcal{F},\, d(g,g')\le\epsilon_m}(X_g-X_{g'})\right] + 12\sum_{j=1}^{m}\int_{\epsilon_{j+1}}^{\epsilon_j}\sqrt{\log N(\epsilon_j,\mathcal{F},d)}\,\mathrm{d}\epsilon \\
&\le \mathbb{E}\left[\sup_{g,g'\in\mathcal{F},\, d(g,g')\le\epsilon_m}(X_g-X_{g'})\right] + 12\sum_{j=1}^{m}\int_{\epsilon_{j+1}}^{\epsilon_j}\sqrt{\log N(\epsilon,\mathcal{F},d)}\,\mathrm{d}\epsilon \\
&= \mathbb{E}\left[\sup_{g,g'\in\mathcal{F},\, d(g,g')\le\epsilon_m}(X_g-X_{g'})\right] + 12\int_{\epsilon_{m+1}}^{\epsilon_1}\sqrt{\log N(\epsilon,\mathcal{F},d)}\,\mathrm{d}\epsilon. \qquad (5.5)
\end{aligned}$$

Let $m\to\infty$, then $\epsilon_m\to 0$, and the first term in (5.5) converges to 0 by sample-continuity. Thus we obtain the bound in (5.1) provided that Dudley's entropy integral exists. □

Remark (Absolute values in suprema). In some cases, we may be interested in the supremum of absolute value. Note that

$$\begin{aligned}
\sup_{f\in\mathcal{F}}|X_f| &= \sup_{f\in\mathcal{F}} X_f + \sup_{f\in\mathcal{F}}(-X_f) - \sup_{f\in\mathcal{F}} X_f \wedge \sup_{f\in\mathcal{F}}(-X_f) \\
&= \sup_{f\in\mathcal{F}} X_f + \sup_{f\in\mathcal{F}}(-X_f) + \inf_{f\in\mathcal{F}} X_f \vee \inf_{f\in\mathcal{F}}(-X_f) \\
&\le \sup_{f\in\mathcal{F}} X_f + \sup_{f\in\mathcal{F}}(-X_f) + \inf_{f\in\mathcal{F}}(X_f\vee(-X_f)) = \sup_{f\in\mathcal{F}} X_f + \sup_{f\in\mathcal{F}}(-X_f) + \inf_{f\in\mathcal{F}}|X_f|.
\end{aligned}$$

Then by applying Theorem 5.3 to both X_f and $-X_f$, we have

$$\begin{aligned}
\mathbb{E}\left[\sup_{f\in\mathcal{F}}|X_f|\right] &\le \mathbb{E}\left[\sup_{f\in\mathcal{F}} X_f\right] + \mathbb{E}\left[\sup_{f\in\mathcal{F}}(-X_f)\right] + \inf_{f\in\mathcal{F}}\mathbb{E}|X_f| \\
&\le 24\int_0^{D/2}\sqrt{\log N(\epsilon,\mathcal{F},d)}\,\mathrm{d}\epsilon + \inf_{f\in\mathcal{F}}\mathbb{E}|X_f|.
\end{aligned}$$

We can also use the chaining rule to construct a tail bound for the supremum of a sub-Gaussian process.

Lemma 5.4 (Adapted from Lemma 3.2 of van de Geer 2000)**.** *Suppose $(\mathcal{F},d)$ and $\{X_f, f\in\mathcal{F}\}$ are the metric space and the stochastic process proposed in Theorem 5.3, the entropy integral in the RHS of* (5.1) *exists, and $\exists f_0\in\mathcal{F}$ such that $X_{f_0}=0$. Then there exist constants $C_0, C_1>0$ depending only on $\mathcal{F}$, such that for all $t>C_0D$, we have*

$$\mathbb{P}\left(\sup_{f\in\mathcal{F}} X_f \ge t\right) \le C_1\exp\left(-\frac{t^2}{C_1^2D^2}\right). \tag{5.6}$$

Proof. We inherit the definition of $\epsilon_j = D2^{-j}, \mathcal{N}_j$ and $\mathcal{H}_j$ from Theorem 3, with the crudest ϵ_0-net being $\mathcal{N}_0=\{f_0\}$. Take C_0 sufficiently large such that

$$t \ge \left(12\sum_{j=1}^{\infty}\epsilon_j\sqrt{2\log|\mathcal{N}_j|}\right)\vee 6D \ge 24D\sqrt{\log\frac{24}{23}}. \tag{5.7}$$

Inspired by (5.2) and (5.3), we choose a sequence $\{\eta_j\}_{j=1}^{\infty}$ such that $\sum_{j=1}^{\infty}\eta_j \le 1$. Then

$$\begin{aligned}\mathbb{P}\left(\sup_{f\in\mathcal{F}} X_f \ge t\right) &\le \mathbb{P}\left(\sum_{j=1}^{\infty}\max_{(g_{j-1},g_j)\in\mathcal{H}_j}\left(X_{g_j}-X_{g_{j-1}}\right)\ge t\sum_{j=1}^{\infty}\eta_j\right)\\ &\le \sum_{j=1}^{\infty}\mathbb{P}\left(\max_{(g_{j-1},g_j)\in\mathcal{H}_j}\left(X_{g_j}-X_{g_{j-1}}\right)\ge \eta_j t\right)\le \sum_{j=1}^{\infty}\exp\left(2\log|\mathcal{N}_j|-\frac{\eta_j^2t^2}{18\epsilon_j^2}\right),\end{aligned} \tag{5.8}$$

where the last inequality follows from (3.7). Now take

$$\eta_j = \frac{6\epsilon_j\sqrt{2\log|\mathcal{N}_j|}}{t}\vee\frac{2^{-j}\sqrt{j}}{4}, \tag{5.9}$$

then by (5.7) we have

$$\sum_{j=1}^{\infty}\eta_j \le \frac{6\sqrt{2}}{t}\sum_{j=1}^{\infty}\epsilon_j\sqrt{\log|\mathcal{N}_j|}+\frac{1}{4}\sum_{j=1}^{\infty}2^{-j}\sqrt{j}\le\frac{1}{2}+\frac{1}{2}=1.$$

Here we use the bound

$$\sum_{j=1}^{\infty}2^{-j}\sqrt{j}\le\sum_{j=1}^{\infty}2^{-j}j=\frac{2^{-1}}{(1-2^{-1})^2}=2. \tag{5.10}$$

By (5.9), we have that $\log|\mathcal{N}_j|\le\frac{\eta_j^2t^2}{72\epsilon_j}$ and $\eta_j\ge\frac{2^{-j}\sqrt{j}}{4}=\frac{\epsilon_j\sqrt{j}}{4D}$, hence

$$\begin{aligned}\mathbb{P}\left(\sup_{f\in\mathcal{F}} X_f\ge t\right) &= \sum_{j=1}^{\infty}\exp\left(2\log|\mathcal{N}_j|-\frac{\eta_j^2t^2}{18\epsilon_j^2}\right)\le\sum_{j=1}^{\infty}\exp\left(-\frac{\eta_j^2t^2}{36\epsilon_j^2}\right)\le\sum_{j=1}^{\infty}\exp\left(-\frac{jt^2}{576D^2}\right)\\ &= \left[1-\exp\left(-\frac{t^2}{576D^2}\right)\right]^{-1}\exp\left(-\frac{t^2}{576D^2}\right)\le 24\exp\left(-\frac{t^2}{576D^2}\right),\end{aligned} \tag{5.11}$$

where the last inequality uses (5.7). Plug in (5.11) to (5.8), then (5.6) holds for $C_1 = 24$, which concludes the proof. □

5.2 Rademacher Complexity

Definition 5.5 (Empirical Rademacher complexity)**.** The *empirical Rademacher complexity* of a function class $\mathcal{F}$ based on a sample $\{x_i\}_{i=1}^n$ is defined as the expected supremum of inner product with independent Rademacher variables $\{\epsilon_j\}_{j=1}^n$:

$$\mathcal{R}(\mathcal{F},x_{1:n}) := \mathbb{E}\left[\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n\epsilon_i f(x_i)\right].$$

Denote by P_n the empirical distribution of $\{x_1,\cdots,x_n\}$. Then we can define norm and inner product on $L^2(P_n)$ space:

$$\|f\|_{P_n}=\left(\frac{1}{n}\sum_{i=1}^n f(x_i)^2\right)^{1/2},\quad \langle f,g\rangle_{P_n}=\frac{1}{n}\sum_{i=1}^n f(x_i)g(x_i).$$

Now we define the process

$$Z_f := \frac{1}{\sqrt{n}}\sum_{i=1}^n\epsilon_i f(x_i),\ f\in\mathcal{F}.$$

By Proposition 1.5, $(Z_f - Z_g)$ is sub-Gaussian with variance proxy $\|f-g\|_{P_n}^2$ for any $f, g \in \mathcal{F}$, namely, $\{Z_f, f \in \mathcal{F}\}$ is sub-Gaussian with respect to $\|\cdot\|_{P_n}$. It is worth noting that $\|\cdot\|_{P_n}$ is possibly a pseudo-metric on $\mathcal{F}$, which means that $\|f\|_{P_n} = 0$ does not necessarily imply $f = 0$. Nonetheless, this slight change does not impact our conclusion, and you can verify that $\{Z_f, f \in \mathcal{F}\}$ is sample-continuous. Using Theorem 5.3, we can establish the connection between Dudley's entropy integral and Rademacher complexity.

Definition 5.6 (Localized empirical Rademacher complexity and critical radius)**.** Suppose we have a function class $\mathcal{F} : \mathcal{X} \to \mathbb{R}$ that is uniformly bounded by b, i.e. $\forall f \in \mathcal{F}, \|f\|_\infty \le b$. The localized empirical Rademacher complexity of a function class $\mathcal{F}$ based on a sample $\{x_i\}_{i=1}^n$ is defined as

$$\mathcal{R}_{\text{loc}}(\delta, \mathcal{F}, x_{1:n}) := \mathbb{E}\left[\sup_{f \in \mathcal{F}: \|f\|_{P_n} \le \delta} \frac{1}{n}\sum_{i=1}^n \epsilon_i f(x_i)\right] = \mathcal{R}(\mathcal{F} \cap B_n(\delta), x_{1:n}),$$

where $\{\epsilon_i\}_{i=1}^n$ are independent Rademacher variables and $B_n(\delta)$ is the closed ball in the norm $\|\cdot\|_{P_n}$ of radius $\delta > 0$ centered at the origin. The empirical critical radius of $\mathcal{F}$ on dataset $\{x_i\}_{i=1}^n$ is defined as the minimum solution smallest positive solution to $\mathcal{R}_{\text{loc}}(\delta, \mathcal{F}, x_{1:n}) \le \delta^2/b$:

$$\widehat{\delta}_n = \min\left\{\delta > 0 : \mathcal{R}_{\text{loc}}(\delta, \mathcal{F}, x_{1:n}) \le \frac{\delta^2}{b}\right\}.$$

Proposition 5.7. *Denote by $B_n(\rho)$ the closed ball in the norm $\|\cdot\|_{P_n}$ of radius $\rho > 0$ centered at the origin. Then the empirical Rademacher complexity of $\mathcal{F}$ satisfies*

$$\mathcal{R}_{\text{loc}}(\rho, \mathcal{F}, x_{1:n}) \le \frac{12}{\sqrt{n}} \int_0^\rho \sqrt{\log N(\epsilon, \mathcal{F}, \|\cdot\|_{P_n})}\, \mathrm{d}\epsilon. \tag{5.12}$$

Proof. Applying Theorem 5.3 to $\left\{Z_f := n^{-1/2}\sum_{i=1}^n \epsilon_i f(x_i), f \in \mathcal{F}\right\}$ immediately concludes the proof. □

5.3 Sub-Gaussian Complexity

Motivation. Let's consider a penalized least square problem. Suppose we have data $\{(x_i, y_i)\}_{i=1}^n \subset \mathcal{X} \times \mathcal{Y}$ collected from

$$y_i = f^*(x_i) + \epsilon_i,\ i = 1, \cdots, n.$$

Given a vector space $\mathcal{F}$ of mappings from $\mathcal{X}$ to $\mathcal{Y}$ with $f^* \in \mathcal{F}$, and let J be seminorm on $\mathcal{F}$. We construct an estimator of f^* from this class by minimizing the regularized risk for some tuning parameter $\lambda \ge 0$:

$$\widehat{f} = \operatorname*{argmin}_{f \in \mathcal{F}} \left\{\frac{1}{n}\sum_{i=1}^n (y_i - f(x_i))^2 + \lambda J(f)\right\}.$$

Recall that we denote by P_n the empirical distribution of $\{x_1, \cdots, x_n\}$. We also abuse the notation $\langle\cdot,\cdot\rangle_{P_n}$ by defining $\langle\cdot,\cdot\rangle_{P_n} : \mathbb{R}^n \times \mathcal{F} \to \mathbb{R}, (z, f) \mapsto \frac{1}{n}\sum_{i=1}^n z_i f(x_i)$. Let $Y = (y_1, \cdots, y_n)^\top$, $\epsilon = (\epsilon_1, \cdots, \epsilon_n)^\top$ be the response and noise vectors. Then (5.10) implies

$$\|Y - \widehat{f}\|_{P_n}^2 + \lambda J(\widehat{f}) \le \|Y - f^*\|_{P_n}^2 + \lambda J(f^*),$$

and we have the basic inequality for $\widehat{f}$:

$$\begin{aligned}
\|\widehat{f} - f^*\|_{P_n}^2 &\le 2\langle \epsilon, \widehat{f} - f^* \rangle_{P_n} + \lambda\big(J(f^*) - J(\widehat{f})\big) \\
&\le 2\big(J(f^*) + J(\widehat{f})\big) \left\langle \epsilon, \frac{\widehat{f} - f^*}{J(\widehat{f}) + J(f^*)} \right\rangle_{P_n} + \lambda\big(J(f^*) - J(\widehat{f})\big) \\
&\le 2\big(J(f^*) + J(\widehat{f})\big) \sup_{J(g)\le 1} \langle \epsilon, g \rangle_{P_n} + \lambda\big(J(f^*) - J(\widehat{f})\big).
\end{aligned}$$

Then we can bound the empirical estimation error by controlling the supremum of an empirical process $\{Z_g := \langle \epsilon, g \rangle_{P_n}\}$ indexed by g. Generally, for a function class $\mathcal{F}$, we call $\sup_{f\in\mathcal{F}} |\langle \epsilon, f \rangle_{P_n}|$ the sub-Gaussian complexity associated with $\mathcal{F}$.

Lemma 5.8 (Adapted from Lemma 8.4 of van de Geer 2000)**.** *Let $\{\epsilon_i, i = 1, \cdots, n\}$ denote independent sub-Gaussian random variables, each having mean zero and variance proxy σ^2. Assume that there exist constants $0 < w < 2$ and $C > 0$ such that for some fixed $x_1, \cdots, x_n$ (which define the empirical norm $\|\cdot\|_{P_n}$),*

$$\log N(\eta, \mathcal{F}, \|\cdot\|_{P_n}) \le C\eta^{-w} \tag{5.13}$$

for sufficiently small $\eta > 0$. Then for any fixed $\rho > 0$, there exists constants $c, c' > 0$, depending only on σ, ρ, C, w such that for all $\gamma > c'$,

$$\sup_{f\in\mathcal{F}\cap B_n(\rho)} \frac{\langle \epsilon, f \rangle_{P_n}}{\|f\|_{P_n}^{1-w/2}} \le \frac{\gamma}{\sqrt{n}}$$

with probability at least $1 - c\exp\left(\frac{\gamma^2}{c^2}\right)$.

Proof. Note that $\frac{\langle \epsilon, f \rangle_{P_n}}{\sqrt{\sigma^2/n}}$ is a sub-Gaussian process with respect to $\|\cdot\|_{P_n}$. For any $0 < \delta < \rho$, the Dudley's entropy integral satisfies

$$\int_0^\delta \sqrt{\log N(\eta, \mathcal{F}, \|\cdot\|_{P_n})}\, \mathrm{d}\eta \le c_0 \delta^{1-w/2} \tag{5.14}$$

for some constant $c_0 > 0$, hence is bounded. By Lemma 5.4, there exists $c_1, c_2 > 0$ depending only on σ, ρ, C, w such that for all $T \ge \frac{c_1\delta}{\sqrt{n}}$, we have

$$\mathbb{P}\left(\sup_{f\in\mathcal{F}\cap B_n(\delta)} \langle \epsilon, f \rangle_{P_n} \ge T \right) \le c_2 \exp\left(-\frac{nT^2}{c_2^2\sigma^2\delta^2} \right).$$

Then for any $T \geq \frac{c_1}{\sqrt{n}} 2^{1-w/2} \rho^{w/2}$, we have

$$\begin{aligned}
\mathbb{P}\left(\sup_{f \in \mathcal{F} \cap B_n(\rho)} \frac{\langle \epsilon, f \rangle_{P_n}}{\|f\|_{P_n}^{1-w/2}} \geq T \right) &= \mathbb{P}\left(\bigcup_{j=1}^{\infty} \left\{ \sup_{f \in \mathcal{F} \cap (B_n(2^{1-j}\rho) \setminus B_n(2^{-j}\rho))} \frac{\langle \epsilon, f \rangle_{P_n}}{\|f\|_{P_n}^{1-w/2}} \geq T \right\} \right) \\
&\leq \sum_{j=1}^{\infty} \mathbb{P}\left(\sup_{f \in \mathcal{F} \cap B_n(2^{1-j}\rho)} \langle \epsilon, f \rangle_{P_n} \geq T (2^{-j}\rho)^{1-w/2} \right) \\
&\leq \sum_{j=1}^{\infty} c_2 \exp\left(-\frac{nT^2 (2^{-j}\rho)^{2-w}}{c_2^2 \sigma^2 (2^{1-j}\rho)^2} \right) = \sum_{j=1}^{\infty} c_2 \exp\left(-\frac{nT^2 2^{jw}}{4c_2^2 \sigma^2 \rho^w} \right) \\
&\leq \sum_{j=1}^{\infty} c_2 \exp\left(-\frac{nT^2 (1 + jw \log 2)}{4c_2^2 \sigma^2 \rho^w} \right) \\
&= c_2 \exp\left(-\frac{nT^2}{4c_2^2 \sigma^2 \rho^w} \right) \frac{\exp\left(-\frac{nT^2 w \log 2}{4c_2^2 \sigma^2 \rho^w} \right)}{1 - \exp\left(-\frac{nT^2 w \log 2}{4c_2^2 \sigma^2 \rho^w} \right)} \\
&\leq c_2 \exp\left(-\frac{nT^2}{4c_2^2 \sigma^2 \rho^w} \right) \frac{\exp\left(-\frac{c_1^2 w \log 2}{c_2^2 2^w \sigma^2} \right)}{1 - \exp\left(-\frac{c_2^2 w \log 2}{c_1^2 2^w \sigma^2} \right)} \leq c \exp\left(-\frac{nT^2}{c^2} \right)
\end{aligned}$$

for some $c > 0$. Set $\gamma = \frac{T}{\sqrt{n}}$, then there exists $c' > 0$ depending only on ρ, σ, C, w such that for any $\gamma \geq c'$,

$$\mathbb{P}\left(\sup_{f \in \mathcal{F} \cap B_n(\rho)} \frac{\langle \epsilon, f \rangle_{P_n}}{\|f\|_{P_n}^{1-w/2}} \geq \frac{\gamma}{\sqrt{n}} \right) \leq c \exp\left(-\frac{\gamma^2}{c^2} \right).$$

Thus we complete the proof. □

This Lemma gives a bound of the empirical process $\{\langle \epsilon, f \rangle_{P_n}, f \in \mathcal{F}\}$. Suppose that $\|f\|_{P_n}$ decays with a rate of $n^{-1/(2+w)}$, then with high probability, the following inequality holds uniformly for all $f \in \mathcal{F}$:

$$\langle \epsilon, f \rangle_{P_n} \lesssim \mathcal{O}\left(\frac{\|f\|_{P_n}^{1-w/2}}{\sqrt{n}} \right) \eqsim \mathcal{O}\left(n^{-\frac{2}{2+w}} \right).$$

Lemma 5.9. *Assume that $\mathcal{F}$ satisfies the entropy bound* (5.13) *for fixed $x_1, \cdots, x_n$, where $0 < w < 2$ and $C > 0$ are constants. Then the empirical critical radius of $\mathcal{F}$ satisfies $\widehat{\delta}_n \leq c_1 n^{-1/(2+w)}$ for a constant c_1.*

Proof. By (5.12) and (5.14), we have

$$\mathcal{R}_{\text{loc}}(\delta, \mathcal{F}, x_{1:n}) \leq \frac{c_0}{\sqrt{n}} \delta^{1-w/2}$$

for some constant $c_0 > 0$. Then the smallest solution $\widehat{\delta}_n$ to $\mathcal{R}_{\text{loc}}(\delta, \mathcal{F}, x_{1:n}) \leq \delta^2/b$ can be upper bounded by

$$\delta^2/b = \frac{c_0}{\sqrt{n}} \delta^{1-w/2} \;\Leftrightarrow\; \delta^{1+w/2} = \frac{c_0 b}{\sqrt{n}},$$

which gives $\widehat{\delta}_n \leq c_1 n^{-1/(2+w)}$ for some constants c_1. □

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